

Last login: Tue Jun 23 09:02:53 on ttys003
adityadave@Adityas-MacBook-Air-3 ~ % psql -d tpch
psql (18.4)
Type "help" for help.

tpch=# SET ROLE alice;

EXPLAIN (ANALYZE, VERBOSE, BUFFERS)

```
select
    s_acctbal,
    s_name,
    n_name,
    p_partkey,
    p_mfgr,
    s_address,
    s_phone,
    s_comment
from
    part,
    supplier,
    partsupp,
    nation,
    region
where
    p_partkey = ps_partkey
    and s_supkey = ps_supkey
    and p_size = 15
    and p_type like '%BRASS'
    and s_nationkey = n_nationkey
    and n_regionkey = region.r_regionkey
    and region.r_name = 'EUROPE'
    and ps_supplycost = (
        select
            min(ps_supplycost)
        from
            partsupp,
            supplier,
            nation,
            region
        where
            p_partkey = ps_partkey
            and s_supkey = ps_supkey
            and s_nationkey = n_nationkey
            and n_regionkey = region.r_regionkey
            and region.r_name = 'EUROPE'
    )
order by
    s_acctbal desc,
    n_name,
    s_name,
    p_partkey
limit 100;
SET
WARNING:  Log 1: Planner Input:
DETAIL:   {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname part
:colnames ("p_partkey" "p_name" "p_mfgr" "p_brand" "p_type" "p_size"
"p_container" "p_retailprice" "p_comment")
}
:rtekind 0
:relid 16551
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname supplier
:colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
:rtekind 0
:relid 16541
:inh true
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals (
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 6
:location -1
}
{FUNCEXP
:funcid 316
:funcresulttype 701
:funcsetset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4

```

```

        :constbyval true
        :constisnull false
        :location -1
        :constvalue 4 { 0 0 0 0 0 0 0 0 }
    }
}
:location -1
}
)
:location -1
}
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStat <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTRY
:alias
{ALIAS
:aliasname n
:colnames <>
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_commen
t")
}
:rtekind 0
:relid 16536
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromC1 true
:securityQuals <>
}
{RANGETBLENTRY
:alias
{ALIAS
:aliasname r
:colnames <>
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:relid 16531
:inh true
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromC1 true
:securityQuals <>
}
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16536
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16531
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
):jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtable 1
}
{RANGETBLREF
:rtable 2
}
)
):quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:varstype 23
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 2
:varattno 4
:varstype 23
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 2
:varattnosyn 4

```

```

        :location -1
    }
}
:location -1
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
)
:location -1
}
{OPEXPR
:opno 98
:opfuncid 67
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 2
:varattno 2
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location -1
}
{CONST
:consttype 25
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
)
:location -1
}
)
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0 0 0 0 0 ]
}
}
:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
}
:location -1
}
}
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname partsupp
:colnames ("ps_partkey" "ps_supkey" "ps_availqty" "ps_supplycost" "p
s_comment")
}
:rtekind 0
:relid 16556
:inh true
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>

```

```

:lateral false
:inFromCl true
:securityQuals (
  {BOOLEXP
    :boolop and
    :args (
      {OPEXP
        :opno 521
        :opfuncid 147
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
          {VAR
            :varno 3
            :varattno 3
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 3
            :varattnosyn 3
            :location -1
          }
          {CONST
            :consttype 23
            :consttypmod -1
            :constcollid 0
            :constlen 4
            :constbyval true
            :constisnull false
            :location -1
            :constvalue 4 [ 100 0 0 0 0 0 0 0 ]
          }
        )
      }
    )
    :location -1
  }
  {SUBLINK
    :subLinkType 0
    :subLinkId 0
    :testexpr <>
    :operName <>
    :subselect
      {QUERY
        :commandType 1
        :querySource 0
        :canSetTag true
        :utilityStmt <>
        :resultRelation 0
        :hasAggs false
        :hasWindowFuncs false
        :hasTargetSRFs false
        :hasSubLinks true
        :hasDistinctOn false
        :hasRecursive false
        :hasModifyingCTE false
        :hasForUpdate false
        :hasRowSecurity true
        :isGroupRTE false
        :isReturn false
        :rteList <>
        :rtable (
          {RANGETBLENTY
            :alias
            {ALIAS
              :aliasname s
              :colnames <>
            }
            :eref
            {ALIAS
              :aliasname s
              :colnames ("s_supkey" "s_name" "s_address" "s_nationkey"
                "s_phone" "s_acctbal" "s_comment")
            }
          }
          :rtekind 0
          :relid 16541
          :inh true
          :relkind r
          :rellockmode 1
          :perminfoindex 1
          :tablesample <>
          :lateral false
          :inFromCl true
          :securityQuals (
            {BOOLEXP
              :boolop and
              :args (
                {OPEXP
                  :opno 674
                  :opfuncid 297
                  :opresulttype 16
                  :opretset false
                  :opcollid 0
                  :inputcollid 0
                  :args (
                    {VAR
                      :varno 1
                      :varattno 6
                      :vartype 701
                      :vartypmod -1
                      :varcollid 0
                      :varnullingrels (b)
                      :varlevelsup 0
                      :varreturningtype 0
                      :varnosyn 1
                      :varattnosyn 6
                      :location -1
                    }
                    {FUNCEXP
                      :funcid 316
                      :funcresulttype 701
                      :funcresultset false
                      :funcvariadic false
                      :funcformat 2
                      :funccollid 0
                      :inputcollid 0
                      :args (
                        {CONST
                          :consttype 23
                          :consttypmod -1
                          :constcollid 0
                          :constlen 4
                          :constbyval true
                          :constisnull false
                          :location -1
                          :constvalue 4 [ 0 0 0 0 0 0 0 0 ]
                        }
                      )
                    }
                  )
                }
              )
            )
          }
          :location -1
        )
      }
    )
    :location -1
  }
  {SUBLINK
    :subLinkType 0
    :subLinkId 0
    :testexpr <>
    :operName <>
    :subselect
      {QUERY
        :commandType 1
        :querySource 0
        :canSetTag true
        :utilityStmt <>
        :resultRelation 0

```

```

:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTY
    :alias
    {ALIAS
      :aliasname n
      :colnames <>
    }
    :eref
    {ALIAS
      :aliasname n
      :colnames ("n_nationkey" "n_name" "n_regionke
        y" "n_comment")
    }
    :rtekind 0
    :relid 16536
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
  {RANGETBLENTY
    :alias
    {ALIAS
      :aliasname r
      :colnames <>
    }
    :eref
    {ALIAS
      :aliasname r
      :colnames ("r_regionkey" "r_name" "r_comment"
    )
    }
    :rtekind 0
    :relid 16531
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 2
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 16536
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 10)
    :insertedCols (b)
    :updatedCols (b)
  }
  {RTEPERMISSIONINFO
    :relid 16531
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
{FROMEXPR
:fromlist (
  {RANGETBLREF
    :rtindex 1
  }
  {RANGETBLREF
    :rtindex 2
  }
)
:quals
{BOOLEXP
:boolop and
:args (
  {OPEXP
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
        :location -1
      }
      {VAR
        :varno 1
        :varattno 4
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 1
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 4
        :location -1
      }
    )
  }
  :location -1
}
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0

```

```

:varnosyn 1
:varattnosyn 3
:location -1
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
}
:location -1
}
{OPEXPR
:opno 98
:opfuncid 67
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 2
:varattno 2
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location -1
}
{CONST
:consttype 25
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 69 85 82 79 8
0 69 ]
}
)
:location -1
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0 0 0 0 0 ]
}
:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
}
:location -1
}
}
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16541
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
}
:quals
{BOOLEXPR
:boolop and
:args (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1

```

```

:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 3
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 3
:varattnosyn 2
:location -1
}
)
:location -1
}
{OPEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{FUNCEXPR
:funcid 316
:funcresulttype 701
:funcsetset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 0 0 0 0 0 0 0 0 ]
}
)
}
:location -1
}
)
:location -1
}
)
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0 0 0 0 0 ]
}
}
:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
)
:location -1
}
)
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname nation
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtekind 0
:relid 16536
:inh true
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref

```

```

        {ALIAS
        :aliasname region
        :colnames ("r_regionkey" "r_name" "r_comment")
        }
:rtelkind 0
:relid 16531
:inh true
:relkind r
:rellockmode 1
:perminfoindex 5
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16551
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10 12 13)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16541
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16556
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16536
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16531
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
{RANGETBLREF
:rtindex 3
}
{RANGETBLREF
:rtindex 4
}
{RANGETBLREF
:rtindex 5
}
)
)
:qual
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 283
}
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 295
}
)
:location 293
}
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 318
}
{VAR
:varno 3
:varattno 2
:vartype 23

```

```

:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 2
:location 330
}
)
:location 328
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 353
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 362
:constvalue 4 [ 15 0 0 0 0 0 0 0 ]
}
)
:location 360
}
{OPEXPR
:opno 1209
:opfuncid 850
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 1
:varattno 5
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 377
}
{CONST
:consttype 25
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 389
:constvalue 10 [ 40 0 0 0 37 66 82 65 83 83 ]
}
)
:location 384
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 4
:location 410
}
{VAR
:varno 4
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location 424
}
)
:location 422
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 3
:location 440
}
{VAR
:varno 5
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5

```

```

:varattnosyn 1
:location 462
}
)
:location 460
}
{OPEXPR
:opno 98
:opfuncid 67
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 5
:varattno 2
:varatype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 2
:location 493
}
{CONST
:consttype 25
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 509
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
)
:location 507
}
{OPEXPR
:opno 670
:opfuncid 293
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 4
:varatype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 4
:location 530
}
{SUBLINK
:sublinktype 4
:sublinkid 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname partsupp
:colnames ("ps_partkey" "ps_suppkey" "ps_availqty" "ps_supplycost" "ps_comment")
}
:rtekind 0
:relid 16556
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals (
{BOOLEXP
:boolop and
:args (
{OPEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 100 0 0 0 0 0 0 0 ]
}
)
}
:location -1
}
}
{SUBLINK
:sublinktype 0
:sublinkid 0
:testexpr <>
:operName <>
:subselect
{QUERY

```

```

:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBENTRY
    :alias
    {ALIAS
      :aliasname s
      :colnames <>
    }
    :eref
    {ALIAS
      :aliasname s
      :colnames ("s_supkey" "s_name" "s_address"
        "s_nationkey" "s_phone" "s_acctbal" "s_c
        omment")
    }
    :rtekind 0
    :relid 16541
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCI true
    :securityQuals (
      {BOOLEXP
        :boolop and
        :args (
          {OPEXP
            :opno 674
            :opfuncid 297
            :opresulttype 16
            :opretset false
            :opcollid 0
            :inputcollid 0
            :args (
              {VAR
                :varno 1
                :varattno 6
                :vartype 701
                :vartypmod -1
                :varcollid 0
                :varnullingrels (b)
                :varlevelsup 0
                :varreturningtype 0
              }
              :varnosyn 1
              :varattnosyn 6
              :location -1
            )
            {FUNCEXP
              :funcid 316
              :funcresulttype 701
              :funcretset false
              :funcvariadic false
              :funcformat 2
              :funccollid 0
              :inputcollid 0
              :args (
                {CONST
                  :consttype 23
                  :consttypmod -1
                  :constcollid 0
                  :constlen 4
                  :constbyval true
                  :constisnull false
                  :location -1
                  :constvalue 4 [ 0 0 0 0 0 0 0 0 ]
                }
              )
            :location -1
          }
        )
      }
    :location -1
  )
  {SUBLINK
    :subLinkType 0
    :subLinkId 0
    :testexpr <>
    :operName <>
    :subselect
    {QUERY
      :commandType 1
      :querySource 0
      :canSetTag true
      :utilityStmt <>
      :resultRelation 0
      :hasAggs false
      :hasWindowFuncs false
      :hasTargetSRFs false
      :hasSubLinks false
      :hasDistinctOn false
      :hasRecursive false
      :hasModifyingCTE false
      :hasForUpdate false
      :hasRowSecurity false
      :hasGroupRTE false
      :isReturn false
      :cteList <>
      :rtable (
        {RANGETBENTRY
          :alias
          {ALIAS
            :aliasname n
            :colnames <>
          }
          :eref
          {ALIAS
            :aliasname n
            :colnames ("n_nationkey" "n_na
              me" "n_regionkey" "n_comment")
          }
          :rtekind 0
          :relid 16536
          :inh true
          :relkind r
          :rellockmode 1
          :perminfoindex 1
          :tablesample <>
          :lateral false
          :inFromCI true
          :securityQuals <>
        }
        {RANGETBENTRY
          :alias
          {ALIAS
            :aliasname r
            :colnames <>
          }
          :eref
          {ALIAS
            :aliasname r
            :colnames ("r_regionkey" "r_na

```

```

        me" "r_comment")
    }
    :rtekind 0
    :reliid 16531
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 2
    :tablesampl <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :reliid 16536
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 10)
    :insertedCols (b)
    :updatedCols (b)
  }
  {RTEPERMISSIONINFO
    :reliid 16531
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
{FROMEXPR
  :fromlist (
    {RANGETBLREF
      :rtindex 1
    }
    {RANGETBLREF
      :rtindex 2
    }
  )
}
:quals
{BOOLEXP
  :boolop and
  :args (
    {OPEXP
      :opno 96
      :opfuncid 65
      :opresulttype 16
      :opretset false
      :opcollid 0
      :inputcollid 0
      :args (
        {VAR
          :varno 1
          :varattno 1
          :vartype 23
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 1
          :location -1
        }
        {VAR
          :varno 1
          :varattno 4
          :vartype 23
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 1
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 4
          :location -1
        }
      )
      :location -1
    }
    {OPEXP
      :opno 96
      :opfuncid 65
      :opresulttype 16
      :opretset false
      :opcollid 0
      :inputcollid 0
      :args (
        {VAR
          :varno 1
          :varattno 3
          :vartype 23
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 3
          :location -1
        }
        {VAR
          :varno 2
          :varattno 1
          :vartype 23
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 2
          :varattnosyn 1
          :location -1
        }
      )
      :location -1
    }
    {OPEXP
      :opno 98
      :opfuncid 67
      :opresulttype 16
      :opretset false
      :opcollid 0
      :inputcollid 100
      :args (
        {VAR
          :varno 2
          :varattno 2
          :vartype 25
          :vartypmod -1
          :varcollid 100
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 2
          :varattnosyn 2
          :location -1
        }
        {CONST
          :consttype 25
          :consttypmod -1
          :constcollid 100
          :constlen -1
        }
      )
    }
  )
}

```

```

                                :constbyval false
                                :constisnull false
                                :location -1
                                :constvalue 10 [ 40 0 0
                                0 69 85 82 79 80 69 ]
                            }
                        }
                    }
                }
            }
        }
    }
}

:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetlist (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0 0 0
0 ]
}
}:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
})
}
}
}
}

:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupByClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

:location -1
}

}

:location -1
}

)

}

)

:rteperminfos (
{RTEPERMISSIONINFO
:relid 16541
:inh true
:requiresPerms 2
:checkedAsUser 0
:selectedCols (b 8 13)
:insertedCols (b)
:updateCols (b)
})
}

:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
})
}
)

quals
{BOOLEXPRESS
:boolop and
:args (
{OEXPR
:opno 96
:opfncid 65
:opresulttype 16
:opretset false
:opcoid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varrattnosyn 1
:location -1
})
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varrattnosyn 2
:location -1
})
}
}

:location -1
}

{OEXPR
:opno 674
:opfncid 297
:opresulttype 16
:opretset false
:opcoid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varrattnosyn 6
:location -1
})
}
}
}

```

```

        {FUNCEXPR
        :funcid 316
        :funcresulttype 701
        :funcresultset false
        :funcvariadic false
        :funcformat 2
        :funccollid 0
        :inputcollid 0
        :args (
        {CONST
        :consttype 23
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 4 [ 0 0 0 0 0 0 0 0 ]
        }
        )
        :location -1
    }
    :location -1
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0 0 0 0 0 ]
}
:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
)
:location -1
}
)
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname supplier
:colnames ("s_supkey" "s_name" "s_address" "s_nationk
ey" "s_phone" "s_acctbal" "s_comment")
}
:rtekind 0
:reld 16541
:inh true
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromC1 true
:securityQuals (
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcoid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 6
:varstype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 6
:location -1
}
{FUNCEXPR
:funcid 316
:funcresulttype 701
:funcresultset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 0 0 0 0 0 0 0 0 ]
}
)
:location -1
}
)
}
:location -1
}
)
:location -1

```

```

}
{SUBLINK
:sublinkType 0
:sublinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname n
:colnames <>
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regio
nkey" "n_comment")
}
:rtekind 0
:relid 16536
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname r
:colnames <>
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comme
nt")
}
:rtekind 0
:relid 16531
:inh true
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16536
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16531
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
)
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 2
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 2
:varattnosyn 4
:location -1
}
}
:location -1
}
)
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16

```

```

:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location -1
  }
  {VAR
    :varno 2
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 1
    :location -1
  }
)
:location -1
}
{OPEXPR
:opno 98
:opfuncid 67
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
  {VAR
    :varno 2
    :varattno 2
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 2
    :location -1
  }
  {CONST
    :consttype 25
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location -1
    :constvalue 10 [ 40 0 0 0 69 85 82 7
      9 88 69 ]
  }
)
:location -1
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {CONST
      :consttype 23
      :consttypmod -1
      :constcollid 0
      :constlen 4
      :constbyval true
      :constisnull false
      :location -1
      :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
    }
    :resno 1
    :resname ?column?
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
}
:location -1
}
}
{RANGETBLENTY
:alias <>
:eref
  {ALIAS
    :aliasname nation
    :colnames ("n_nationkey" "n_name" "n_regionkey" "n_com
      ment")
  }
:rtekind 0
:relid 16536
:inh true
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
  {ALIAS

```

```

        :aliasname region
        :colnames ("r_regionkey" "r_name" "r_comment")
    }
    :rtekind 0
    :relid 16531
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 4
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
}
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 16556
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9 11)
    :insertedCols (b)
    :updatedCols (b)
  }
  {RTEPERMISSIONINFO
    :relid 16541
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 11)
    :insertedCols (b)
    :updatedCols (b)
  }
  {RTEPERMISSIONINFO
    :relid 16536
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 10)
    :insertedCols (b)
    :updatedCols (b)
  }
  {RTEPERMISSIONINFO
    :relid 16531
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
{FROMEXPR
  :fromlist (
    {RANGETBLREF
      :rtindex 1
    }
    {RANGETBLREF
      :rtindex 2
    }
    {RANGETBLREF
      :rtindex 3
    }
    {RANGETBLREF
      :rtindex 4
    }
  )
}
:quals
{BOOLEXP
  :boolop and
  :args (
    {OPEXP
      :opno 96
      :opfuncid 65
      :opresulttype 16
      :opretset false
      :opcollid 0
      :inputcollid 0
      :args (
        {VAR
          :varno 1
          :varattno 1
          :vartype 23
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 1
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 1
          :location 812
        }
        {VAR
          :varno 1
          :varattno 1
          :vartype 23
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 1
          :location 824
        }
      )
    }
    :location 822
  )
}
{OPEXP
  :opno 96
  :opfuncid 65
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
    {VAR
      :varno 2
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 2
      :varattnosyn 1
      :location 863
    }
    {VAR
      :varno 1
      :varattno 2
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 2
      :location 875
    }
  )
}
:location 873
}
{OPEXP

```

```

:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 2
    :varattno 4
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 4
    :location 914
  }
  {VAR
    :varno 3
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 1
    :location 928
  }
)
:location 926
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 3
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 3
    :location 968
  }
  {VAR
    :varno 4
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 4
    :varattnosyn 1
    :location 982
  }
)
:location 980
}
{OPEXPR
:opno 98
:opfuncid 67
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
  {VAR
    :varno 4
    :varattno 2
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 4
    :varattnosyn 2
    :location 1029
  }
  {CONST
    :consttype 25
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location 1045
    :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
  }
)
:location 1043
}
}
:location 859
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {AGGREF
      :aggfnoid 2136
      :aggtype 701
      :aggcollid 0
      :inputcollid 0
      :aggtranstype 0
      :aggargtypes (o 701)
      :aggdirectargs <>
    }
    :args (
      {TARGETENTRY
        :expr
        {VAR
          :varno 1
          :varattno 4
          :vartype 701
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 4
          :location 599
        }
      }
      :resno 1
      :resname <>
      :ressortgroupref 0
      :resorigtbl 0
      :resorigcol 0
      :resjunk false
    )
  }
)

```

```

    )
    :aggorder <>
    :aggdistinct <>
    :aggfilter <>
    :aggstar false
    :aggvariadic false
    :aggkind n
    :aggpresorted false
    :agglevelsup 0
    :aggsplit 0
    :aggno -1
    :aggtransno -1
    :location 595
  }
  :resno 1
  :resname min
  :ressortgrouppref 0
  :resorigtbl 0
  :resorigcol 0
  :resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:location 546
}
)
:location 544
}
)
:location 314
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetlist (
  {TARGETENTRY
  :expr
  {VAR
  :varno 2
  :varattno 6
  :vartype 701
  :vartypmod -1
  :varcollid 0
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 2
  :varattnosyn 6
  :location 51
  }
  :resno 1
  :resname s_acctbal
  :ressortgrouppref 1
  :resorigtbl 16541
  :resorigcol 6
  :resjunk false
  }
  {TARGETENTRY
  :expr
  {VAR
  :varno 2
  :varattno 2
  :vartype 25
  :vartypmod -1
  :varcollid 100
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 2
  :varattnosyn 2
  :location 70
  }
  :resno 2
  :resname s_name
  :ressortgrouppref 3
  :resorigtbl 16541
  :resorigcol 2
  :resjunk false
  }
  {TARGETENTRY
  :expr
  {VAR
  :varno 4
  :varattno 2
  :vartype 25
  :vartypmod -1
  :varcollid 100
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 4
  :varattnosyn 2
  :location 86
  }
  :resno 3
  :resname n_name
  :ressortgrouppref 2
  :resorigtbl 16536
  :resorigcol 2
  :resjunk false
  }
  {TARGETENTRY
  :expr
  {VAR
  :varno 1
  :varattno 1
  :vartype 23
  :vartypmod -1
  :varcollid 0
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 1
  :varattnosyn 1
  :location 102
  }
  :resno 4
  :resname p_partkey
  :ressortgrouppref 4
  :resorigtbl 16551
  :resorigcol 1
  :resjunk false
  }
  {TARGETENTRY
  :expr
  {VAR

```

```

:varno 1
:varattno 3
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 121
}
:resno 5
:resname p_mfgr
:ressortgroupref 0
:resorigtbl 16551
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 3
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 3
:location 137
}
:resno 6
:resname s_address
:ressortgroupref 0
:resorigtbl 16541
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 5
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 5
:location 156
}
:resno 7
:resname s_phone
:ressortgroupref 0
:resorigtbl 16541
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 7
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 7
:location 173
}
:resno 8
:resname s_comment
:ressortgroupref 0
:resorigtbl 16541
:resorigcol 7
:resjunk false
}
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
{SORTGROUPCLAUSE
:tSortGroupRef 1
:eqop 670
:sortop 674
:reverse_sort true
:nulls_first true
:hashable true
}
{SORTGROUPCLAUSE
:tSortGroupRef 2
:eqop 98
:sortop 664
:reverse_sort false
:nulls_first false
:hashable true
}
{SORTGROUPCLAUSE
:tSortGroupRef 3
:eqop 98
:sortop 664
:reverse_sort false
:nulls_first false
:hashable true
}
{SORTGROUPCLAUSE
:tSortGroupRef 4
:eqop 96
:sortop 97
:reverse_sort false
:nulls_first false
:hashable true
}
)
:limitOffset <>
:limitCount
{FUNCEXP
:funcid 481
:funcresulttype 20
:funcresultset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
(CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true

```

```

:constisnull false
:location 1153
:constvalue 4 [ 100 0 0 0 0 0 0 ]
}
}
:location -1
}
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

WARNING: max_parallel_hazard_walker: FuncExpr funcid=316 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: max_parallel_hazard_walker: FuncExpr funcid=316 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: max_parallel_hazard_walker: FuncExpr funcid=316 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: max_parallel_hazard_walker: FuncExpr funcid=481 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: max_parallel_hazard_walker: FuncExpr funcid=316 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: max_parallel_hazard_walker: FuncExpr funcid=316 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: max_parallel_hazard_walker: FuncExpr funcid=316 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: max_parallel_hazard_walker: FuncExpr funcid=316 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: Log 2: After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname n
:colnames <>
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtekind 0
:relid 16536
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname r
:colnames <>
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:relid 16531
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16536
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16531
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
)
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{PARAM
:paramkind 1
:paramid 2
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
}
}
}

```

```
)
:location -1
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location -1
  }
  {VAR
    :varno 2
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 1
    :location -1
  }
)
:location -1
}
{OEXPR
:opno 98
:opfuncid 67
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
  {VAR
    :varno 2
    :varattno 2
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 2
    :location -1
  }
  {CONST
    :consttype 25
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location -1
    :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
  }
)
:location -1
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 2 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_rels (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 2 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_rels (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: max_parallel_hazard_walker: Param paramid=2 in_safe_list=0 → reason=outer_reference
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 2 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_rels (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 2 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_rels (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = false | reason = PROPARALLEL_RESTRICTED
WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 98 :opfuncid 67 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({VAR :varno 2 :varattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1} :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 2) :required_rels (b 2) :incompatible_rels (b) :outer_rels (b 2) :left_rels (b 2) :right_rels (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1994 2095) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_rels (b 2) :em_is_const false :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1} :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]} :em_rels (b) :em_is_const true :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 98 :opfuncid 67 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({VAR :varno 2 :varattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1} :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 2) :required_rels (b 2) :incompatible_rels (b) :outer_rels (b) :left_rels (b 2) :right_rels (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1994 2095) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_rels (b 2) :em_is_const false :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1} :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]} :em_rels (b) :em_is_const true :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL:
:commandType 1
```

```

:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:table (
  {RANGETBENTRY
  :alias
  {ALIAS
  :aliasname n
  :colnames <>
  }
  :eref
  {ALIAS
  :aliasname n
  :colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
  }
  :rtekind 0
  :relid 16536
  :inh false
  :relkind r
  :rellockmode 1
  :perminfoindex 1
  :tablesample <>
  :lateral false
  :inFromCl true
  :securityQuals <>
  }
  {RANGETBENTRY
  :alias
  {ALIAS
  :aliasname r
  :colnames <>
  }
  :eref
  {ALIAS
  :aliasname r
  :colnames ("r_regionkey" "r_name" "r_comment")
  }
  :rtekind 0
  :relid 16531
  :inh false
  :relkind r
  :rellockmode 1
  :perminfoindex 2
  :tablesample <>
  :lateral false
  :inFromCl true
  :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
  :relid 16536
  :inh true
  :requiredPerms 2
  :checkAsUser 0
  :selectedCols (b 8 10)
  :insertedCols (b)
  :updatedCols (b)
  }
  {RTEPERMISSIONINFO
  :relid 16531
  :inh true
  :requiredPerms 2
  :checkAsUser 0
  :selectedCols (b 8 9)
  :insertedCols (b)
  :updatedCols (b)
  }
)
:jointree
{FROMEXPR
:fromlist (
  {RANGETBLREF
  :rtindex 1
  }
  {RANGETBLREF
  :rtindex 2
  }
)
:quals (
  {OEXPR
  :opno 96
  :opfuncid 65
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
    {VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location -1
    }
    {VAR
    :varno 2
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 1
    :location -1
    }
  )
  :location -1
  }
  {OEXPR
  :opno 98
  :opfuncid 67
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 100
  :args (
    {VAR
    :varno 2
    :varattno 2
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
  }
  )
  :location -1
  }
)

```

```
:varattnosyn 2
:location -1
}
{CONST
:consttype 25
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:exp (
{VAR
:varno 1
:varattno 1
:varstype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returninglist <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len 1
}
)
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :varstype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
{VAR :varno 1 :varattno 1 :varstype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
)
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :varstype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
{VAR :varno 1 :varattno 1 :varstype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 98 :opfncid 67 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({VAR :varno 2 :varattno 2 :varstype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 2) :required_rels (b 2) :incompatible_rels (b) :outer_rels (b 2) :right_rels (b) :orclause <> :rinfo :serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1994 2095) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :varstype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_rels (b 2) :em_is_const false :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 98 :opfncid 67 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({VAR :varno 2 :varattno 2 :varstype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 2) :required_rels (b 2) :incompatible_rels (b) :outer_rels (b 2) :right_rels (b) :orclause <> :rinfo :serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1994 2095) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :varstype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_rels (b 2) :em_is_const false :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :varstype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1}
)
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 2 :varattno 1 :varstype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1}
) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 96 :opfncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 3 :varstype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {VAR :varno 2 :varattno 1 :varstype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1}) :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_rels (b 1 2) :required_rels (b 1 2) :incompatible_rels (b) :outer_rels (b 1) :right_rels (b 2) :orclause <> :rinfo :serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec 0.005 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 3 :varstype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} :em_rels (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 1 :varstype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} :em_rels (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 96 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 96 :right_hashoperator 96})
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 96 :opfncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 3 :varstype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {VAR :varno 2 :varattno 1 :varstype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1}) :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_rels (b 1 2) :required_rels (b 1 2) :incompatible_rels (b) :outer_rels (b 1) :right_rels (b 2) :orclause <> :rinfo :serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec 0.005 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 3 :varstype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} :em_rels (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 1 :varstype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} :em_rels (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 96 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 96 :right_hashoperator 96}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :varstype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
)
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :varstype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :varstype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
)
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:ctelist <>
:rttable (
{RANGELEFTENTRY
```

```

:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_supplekey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
:rtekind 0
:relid 16541
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals (
{OPEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varrttno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
){CONST
:consttype 701
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 0 0 0 0 0 0 0 ]
}
}
:location -1
}
){ALTERNATIVESUBPLAN
:subplans (
{SUBPLAN
:subLinkType 0
:testexpr <>
:paramids <>
:plan_id 1
:plan_name SubPlan\ 1
:firstColType 2278
:firstColTypeMod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam (i 2)
:args (
{VAR
:varno 1
:varrttno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
)
:startup_cost 0
:per_call_cost 41
}
{SUBPLAN
:subLinkType 2
:testexpr
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varrttno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
){PARAM
:paramkind 1
:paramid 3
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
}
:location -1
}
):paramids (i 3)
:plan_id 2
:plan_name SubPlan\ 2
:firstColType 23
:firstColTypeMod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 42.012499999999999999
:per_call_cost 0.0025
}
}
}
}
}
):rteperminfos (
(RTEPERMISSIONINFO
:relid 16541
:inh true

```

```

:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
:RANGETBLREF
:rtindex 1
)
}
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{PARAM
:paramkind 1
:paramid 1
:paramtype 23
:paratypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
{OEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{CONST
:consttype 701
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 0 0 0 0 0 0 0 ]
}
)
:location -1
}
)
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:nonConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

```

```

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 674 :opfuncid 297 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 701 :consttypmod -1 :constcollid 0 :constlen 8 :constbyval true :constisnull false :location -1 :constvalue 8 [ 0 0 0 0 0 0 0 0 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0} {RESTRICTINFO :clause {OEXPR :opno 674 :opfuncid 297 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} :startup_cost 0 :per_call_cost 41} {SUBPLAN :sublinktype 2 :testexpr {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} {PARAM :paramkind 1 :paramid 3 :paramtype 23 :paratypmod -1 :paramcollid 0 :location -1}) :location -1} :paramids (i 3) :plan_id 2 :plan_name SubPlan 1 :firstColType 23 :firstColCollation 0 :useHashTable true :unknownEqFalse true :parallel_safe false :setParam <> :parParam <> :args <> :startup_cost 42.012499999999996 :per_call_cost 0.0025}) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0} {RESTRICTINFO :clause {OEXPR :opno 674 :opfuncid 297 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 701 :consttypmod -1 :constcollid 0 :constlen 8 :constbyval true :constisnull false :location -1 :constvalue 8 [ 0 0 0 0 0 0 0 0 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof true :has_volatile 0 :security_level 1 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 4 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0} {RESTRICTINFO :clause {OEXPR :opno 674 :opfuncid 297 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} :startup_cost 0 :per_call_cost 41} {SUBPLAN :sublinktype 2 :testexpr {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} {PARAM :paramkind 1 :paramid 3 :paramtype 23 :paratypmod -1 :paramcollid 0 :location -1}) :location -1} :paramids (i 3) :plan_id 2 :plan_name SubPlan 1 :firstColType 23 :firstColCollation 0 :useHashTable true :unknownEqFalse true :parallel_safe false :setParam <> :parParam <> :args <> :startup_cost 42.012499999999996 :per_call_cost 0.0025}) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_rels (b 1) :em_is_const false :em_datatype 23 :em_domain {JOINDOMAIN :jd_rels (b 1)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 1 :paramtype 23 :paratypmod -1 :paramcollid 0 :location -1} :em_rels (b 1) :em_is_const true :em_datatype 23 :em_domain {JOINDOMAIN :jd_rels (b 1)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0}
WARNING: max_parallel_hazard_walker: SubPlan plan_id=1 parallel_safe=0 > reason=correlated_subplan
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 674 :opfuncid 297 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 701 :consttypmod -1 :constcollid 0 :constlen 8 :constbyval true :constisnull false :location -1 :constvalue 8 [ 0 0 0 0 0 0 0 0 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0} {RESTRICTINFO :clause {OEXPR :opno 674 :opfuncid 297 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} :startup_cost 0 :per_call_cost 41} {SUBPLAN :sublinktype 2 :testexpr {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} {PARAM :paramkind 1 :paramid 3 :paramtype 23 :paratypmod -1 :paramcollid 0 :location -1}) :location -1} :paramids (i 3) :plan_id 2 :plan_name SubPlan 1 :firstColType 23 :firstColCollation 0 :useHashTable true :unknownEqFalse true :parallel_safe false :setParam <> :parParam <> :args <> :startup_cost 42.012499999999996 :per_call_cost 0.0025}) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0} {RESTRICTINFO :clause {OEXPR :opno 674 :opfuncid 297 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} :startup_cost 0 :per_call_cost 41} {SUBPLAN :sublinktype 2 :testexpr {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} {PARAM :paramkind 1 :paramid 3 :paramtype 23 :paratypmod -1 :paramcollid 0 :location -1}) :location -1} :paramids (i 3) :plan_id 2 :plan_name SubPlan 1 :firstColType 23 :firstColCollation 0 :useHashTable true :unknownEqFalse true :parallel_safe false :setParam <> :parParam <> :args <> :startup_cost 42.012499999999996 :per_call_cost 0.0025}) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_rels (b 1) :em_is_const false :em_datatype 23 :em_domain {JOINDOMAIN :jd_rels (b 1)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 1 :paramtype 23 :paratypmod -1 :paramcollid 0 :location -1} :em_rels (b 1) :em_is_const true :em_datatype 23 :em_domain {JOINDOMAIN :jd_rels (b 1)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hashoperator 0 :right_hashoperator 0}

```

```

2 :firstColType 23 :firstColTypmod -1 :firstColCollation 0 :useHashTable true :knownEq false true :parallel_safe true :setParam <> :parParam <> :args <> :startup_cost 42.012499999999996 :per_call_co
st 0.0025))} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :require
d_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mer
geopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator
0} {RESTRICINFO :clause {OEXPR :opno 674 :opfncid 297 :opresultype 16 :opreset false :opcoid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnulling
rels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 701 :consttypmod -1 :constcollid 0 :constbyval true :constisnull false :location -1 :co
nstvalue ({ 0 0 0 0 0 0 })} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof true :has_volatile 0 :security_level 1 :num_base_rel
s 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 4 :eval_cost.startup -1 :eval_cost.per_tuple 0
:norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_has
heqoperator 0 :right_hasheqoperator 0} {RESTRICINFO :clause {OEXPR :opno 96 :opfncid 65 :opresultype 16 :opreset false :opcoid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :vartype 23 :varty
pmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 1 :paramtype 23 :paramtypmod -1 :paramcollid 0 :locatio
n -1} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof true :has_volatile 2 :security_level 1 :num_base_rels 1 :clause_relds (b 1) :r
equired_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec
-1 :mergeopfamilies {o 1976} :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :va
rattnosyn 1 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :par
amid 1 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1)}} :outer_is_left false :has
hjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0)} | parallel_safe = false | reason = PROPARALLEL_RESTRICTED
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL:      QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable {
(RANGETBENTRY
:alias
{ALIAS
:aliasname n
:colnames <>
}
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
}
:rtekind 0
:reldid 16536
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesampl <>
:lateral false
:inFromC1 true
:securityQuals <>
}
(RANGETBENTRY
:alias
{ALIAS
:aliasname r
:colnames <>
}
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
}
:rtekind 0
:reldid 16531
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesampl <>
:lateral false
:inFromC1 true
:securityQuals <>
}
)
:rterpermfos {
(RTEPERMISSIONINFO
:reldid 16536
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatecols (b)
}
(RTEPERMISSIONINFO
:reldid 16531
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatecols (b)
}
}
)
:jointree {
FROMEXPR
:fromlist {
{RANGETBLREF
:rtdindex 1
}
{RANGETBLREF
:rtdindex 2
}
}
}
quals {
{OEXPR
:opno 96
:opfncid 65
:opresultype 16
:opreset false
:opcoid 0
:inputcollid 0
:args {
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{PARAM
:paramkind 1
:paramid 5
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
}
}
:location -1
}
}
{OEXPR

```

```

:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  (VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location -1
  )
  (VAR
    :varno 2
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 1
    :location -1
  )
)
:location -1
)
(OEXPR
  :opno 98
  :opfuncid 67
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 100
  :args (
    (VAR
      :varno 2
      :varattno 2
      :vartype 25
      :vartypmod -1
      :varcollid 100
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 2
      :varattnosyn 2
      :location -1
    )
    (CONST
      :consttype 25
      :consttypmod -1
      :constcollid 100
      :constlen -1
      :constbyval false
      :constisnull false
      :location -1
      :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
    )
  )
  :location -1
)
)
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetlist <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
)
}

WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 5 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b 1) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_rels (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 5 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_rels (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: max_parallel_hazard_walker: Param paramid=5 in safe_list=0 → reason=outer_reference
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 5 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b 1) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_rels (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 5 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_rels (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = false | reason = PROPAGALLEL_RESTRICTED
WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {OEXPR :opno 98 :opfuncid 67 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({VAR :varno 2 :varattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 2) :required_rels (b 2) :incompatible_rels (b) :outer_rels (b) :left_rels (b 2) :right_rels (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1994 2095) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_rels (b 2) :em_is_const false :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :em_rels (b) :em_is_const true :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause {OEXPR :opno 98 :opfuncid 67 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({VAR :varno 2 :varattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 2) :required_rels (b 2) :incompatible_rels (b) :outer_rels (b) :left_rels (b 2) :right_rels (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1994 2095) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_rels (b 2) :em_is_const false :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :em_rels (b) :em_is_const true :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPAGALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = (VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1)
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1) | parallel_safe = true | reason = PROPAGALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <> | parallel_safe = true | reason = PROPAGALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL:
  :QUERY
  :commandType 1
  :querySource 0
  :canSetTag true
  :utilityStmt <>
  :resultRelation 0

```

```

:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBENTRY
    :alias
      {ALIAS
        :aliasname n
        :colnames <>
      }
    :eref
      {ALIAS
        :aliasname n
        :colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
      }
    :rtekind 0
    :relid 16536
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
  {RANGETBENTRY
    :alias
      {ALIAS
        :aliasname r
        :colnames <>
      }
    :eref
      {ALIAS
        :aliasname r
        :colnames ("r_regionkey" "r_name" "r_comment")
      }
    :rtekind 0
    :relid 16531
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 2
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 16536
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 0 10)
    :insertedCols (b)
    :updatedCols (b)
  }
  {RTEPERMISSIONINFO
    :relid 16531
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 0 9)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
{FROMEXPR
  :fromlist (
    {RANGETBLREF
      :rtindex 1
    }
    {RANGETBLREF
      :rtindex 2
    }
  )
}
:quals (
  {OPEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 1
        :varattno 3
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 3
        :location -1
      }
      {VAR
        :varno 2
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 1
        :location -1
      }
    )
    :location -1
  }
  {OPEXPR
    :opno 98
    :opfuncid 67
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 100
    :args (
      {VAR
        :varno 2
        :varattno 2
        :vartype 25
        :vartypmod -1
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 2
        :location -1
      }
    )
  }
  {CONST

```

```
:conststype 25
:consttypemod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
}
:location -1
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 1
      :vartype 23
      :vartypemod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 1
      :location -1
    }
    :resno 1
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
{VAR :varno 1 :varattno 3 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
} {VAR :varno 1 :varattno 3 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1}) | parallel_safe = true | reason = PROP
ARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 98 :opfncid 67 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({VAR :varno 2 :varattno 2 :vartype 25
:vartypemod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 25 :consttypemod -1 :constcollid 100 :constlen -1 :constbyval
false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproo
f false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause <> :rinfo
_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1994 2095) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 25 :vartypemod
-1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 25 :em_jdomain {JOI
NDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 25 :consttypemod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 4
0 0 0 69 85 82 79 80 69 ]} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1
:right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 98 :opfncid 67 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({VAR :varno 2 :varattno 2 :vartype 25
:vartypemod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 25 :consttypemod -1 :constcollid 100 :constlen -1 :constbyva
l false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproo
f false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause <> :rinfo
_o_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1994 2095) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 25 :vartypemod
-1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 25 :em_jdomain {JOI
NDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 25 :consttypemod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 4
0 0 0 69 85 82 79 80 69 ]} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -
1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 2 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1}
)
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 2 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 96 :opfncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 3 :vartype 23 :va
rtypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {VAR :varno 2 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (
b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1}) :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false
:has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b 1 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b 2) :orclause <> :rinfo
_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec 0.005 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 3 :vartype 23 :vartypemod -1
:varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAI
N :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattno
syn 1 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 96 :left_bucketsize -1 :rig
ht_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 96 :right_hasheqoperator 96})
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 96 :opfncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 3 :vartype 23 :v
artypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {VAR :varno 2 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels
(b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1}) :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false
:has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b 1 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b 2) :orclause <> :rinfo
_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec 0.005 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 3 :vartype 23 :vartypemod -1
:varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMA
IN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattno
syn 1 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 96 :left_bucketsize -1 :rig
ht_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 96 :right_hasheqoperator 96}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
)
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}
)
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetFag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:ctelist <>
:rttable (
  {RANGETABLEENTRY
    :alias
    {ALIAS
      :aliasname s
      :colnames <>
    }
  }
)
```

```

    }
:eref
{ALIAS
:aliasname s
:colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
:rtekind 0
:reliid 16541
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals (
{OPEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:varstype 701
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{CONST
:consttype 701
:conststypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 0 0 0 0 0 0 0 ]
}
)
:location -1
}
{ALTERNATIVESUBPLAN
:subplans (
{SUBPLAN
:subLinkType 0
:testexpr <>
:paramids <>
:plan_id 4
:plan_name SubPlan\ 4
:firstColType 2278
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam (i 5)
:args (
{VAR
:varno 1
:varattno 4
:varstype 23
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
)
:startup_cost 0
:per_call_cost 41
}
{SUBPLAN
:subLinkType 2
:testexpr
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 4
:varstype 23
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
{PARAM
:paramkind 1
:paramid 6
:paramtype 23
:paramstypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
:paramids (i 6)
:plan_id 5
:plan_name SubPlan\ 5
:firstColType 23
:firstColTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 42.012499999999996
:per_call_cost 0.0025
}
)
)
)
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:reliid 16541
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 13)
:insertedCols (b)

```

```

      :updatedCols (b)
    }
  }
}
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals (
{OEXPR
:opno 674
:opfncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:varstype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
){CONST
:consttype 701
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 0 0 0 0 0 0 0 ]
}
)
:location -1
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:varstype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn
:varattnosyn 1
:location -1
}
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:tmt_location -1
:tmt_len -1
}

```

```

WARNING: is_parallel_safe: INPUT qual = ({RESTRICINFO :clause {OEXPR :opno 674 :opfncid 297 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :varstype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 701 :consttypmod -1 :constcollid 0 :constlen 8 :constbyval true :constisnull false :location -1 :constvalue 8 [ 0 0 0 0 0 0 0 0 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b) :right_rels (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} {RESTRICINFO :clause {ALTERNATIVESUBPLAN :subplans ({SUBPLAN :subLinkType 0 :testexpr <> :paramids <> :plan_id 4 :plan_name SubPlan 4 :firstCollType 2278 :firstCollTypmod -1 :firstCollCollation 0 :useHashTable false :unknownEqFalse true :parallel_safe false :setParam <> :parParam (i 5) :args ({VAR :varno 1 :varattno 4 :varstype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1} {PARAM :paramkind 1 :paramid 6 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1} :paramids (i 6) :plan_id 5 :plan_name SubPlan 5 :firstCollType 23 :firstCollTypmod -1 :firstCollCollation 0 :useHashTable true :unknownEqFalse true :parallel_safe true :setParam <> :parParam <> :args <> :startup_cost 42.012499999999996 :per_call_cost 0.0025}) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b) :right_rels (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} {RESTRICINFO :clause {OEXPR :opno 674 :opfncid 297 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :varstype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 701 :consttypmod -1 :constcollid 0 :constlen 8 :constbyval true :constisnull false :location -1 :constvalue 8 [ 0 0 0 0 0 0 0 0 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof true :has_volatile 0 :security_level 1 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b) :right_rels (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = false | reason = PROPAGATED_RESTRICTED
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :varstype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {CONST :consttype 701 :consttypmod -1 :constcollid 0 :constlen 8 :constbyval true :constisnull false :location -1 :constvalue 8 [ 0 0 0 0 0 0 0 0 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b) :right_rels (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPAGATED_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL:
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
}

```

```

:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBENTRY
    :alias
    {ALIAS
      :aliasname n
      :colnames <>
    }
    :eref
    {ALIAS
      :aliasname n
      :colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
    }
    :rtekind 0
    :relid 16536
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
  {RANGETBENTRY
    :alias
    {ALIAS
      :aliasname r
      :colnames <>
    }
    :eref
    {ALIAS
      :aliasname r
      :colnames ("r_regionkey" "r_name" "r_comment")
    }
    :rtekind 0
    :relid 16531
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 2
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 16536
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 10)
    :insertedCols (b)
    :updatedCols (b)
  }
  {RTEPERMISSIONINFO
    :relid 16531
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
{FROMEXPR
  :fromlist (
    {RANGETBLREF
      :rtindex 1
    }
    {RANGETBLREF
      :rtindex 2
    }
  )
}
:quals (
  {OEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
        :location -1
      }
      {PARAM
        :paramkind 1
        :paramid 7
        :paramtype 23
        :paramtypmod -1
        :paramcollid 0
        :location -1
      }
    )
    :location -1
  }
  {OEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 1
        :varattno 3
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 3
        :location -1
      }
      {VAR
        :varno 2
        :varattno 1
        :vartype 23
        :vartypmod -1

```

```

:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
}
:location -1
}
{OEXPR
:opno 98
:opfuncid 67
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 2
:varattno 2
:varatype 25
:varitypm 1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location -1
}
{CONST
:consttype 25
:constitypm 1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
}
:location -1
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING: is_parallel_safe: INPUT qual = ({(RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({(VAR :varno 1 :varattno 1 :varatype 23 :varitypm 1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) (PARAM :paramkind 1 :paramid 7 :paramtype 23 :paramitypm 1 :paramcollid 0 :location -1) :location -1}) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :varatype 23 :varitypm 1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_rels (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 7 :paramtype 23 :paramitypm 1 :paramcollid 0 :location -1} :em_rels (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({(RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({(VAR :varno 1 :varattno 1 :varatype 23 :varitypm 1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) (PARAM :paramkind 1 :paramid 7 :paramtype 23 :paramitypm 1 :paramcollid 0 :location -1) :location -1}) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :varatype 23 :varitypm 1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_rels (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 7 :paramtype 23 :paramitypm 1 :paramcollid 0 :location -1} :em_rels (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = false | reason = PROPARALLEL_RESTRICTED
WARNING: is_parallel_safe: INPUT qual = ({(RESTRICTINFO :clause {OEXPR :opno 98 :opfuncid 67 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({(VAR :varno 2 :varattno 2 :varatype 25 :varitypm 1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1) (CONST :consttype 25 :constitypm 1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]) :location -1}) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 2) :required_rels (b 2) :incompatible_rels (b) :outer_rels (b) :left_rels (b 2) :right_rels (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1994 2095) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :varatype 25 :varitypm 1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_rels (b 2) :em_is_const false :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 25 :constitypm 1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]) :em_rels (b) :em_is_const true :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({(RESTRICTINFO :clause {OEXPR :opno 98 :opfuncid 67 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({(VAR :varno 2 :varattno 2 :varatype 25 :varitypm 1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1) (CONST :consttype 25 :constitypm 1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]) :location -1}) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 2) :required_rels (b 2) :incompatible_rels (b) :outer_rels (b) :left_rels (b 2) :right_rels (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1994 2095) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :varatype 25 :varitypm 1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_rels (b 2) :em_is_const false :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 25 :constitypm 1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]) :em_rels (b) :em_is_const true :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({(VAR :varno 2 :varattno 1 :varatype 23 :varitypm 1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1)
WARNING: is_parallel_safe: OUTPUT qual = ({(VAR :varno 2 :varattno 1 :varatype 23 :varitypm 1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1
)) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {
QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:table (
{RANGELEFTENTRY
:alias
{ALIAS
:aliasname n
:colnames <>
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
}
:rtkind 0

```

```

:relid 16536
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname r
:colnames <>
}
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:relid 16531
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16536
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 0 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16531
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 0 9)
:insertedCols (b)
:updatedCols (b)
}
)
):jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rindex 1
}
{RANGETBLREF
:rindex 2
}
)
):quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:varstype 23
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
{VAR
:varno 2
:varattno 1
:varstype 23
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
)
:location -1
}
{OEXPR
:opno 98
:opfuncid 67
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 2
:varattno 2
:varstype 25
:varstypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location -1
}
{CONST
:consttype 25
:conststypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
)
:location -1
}
)
)
):mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:varstype 23
:varstypmod -1

```

```

:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) (VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1)})
WARNING: is_parallel_safe: OUTPUT qual = ({(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) (VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1)}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({(RESTRICTINFO :clause {OEXPR :opno 98 :opfuncid 67 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({(VAR :varno 2 :varattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1) (CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 2) :required_rels (b 2) :incompatible_rels (b) :outer_rels (b) :left_rels (b 2) :right_rels (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1994 2095) :left_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 2 :varattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1) :em_rels (b 2) :em_is_const false :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}) :right_em (EQUIVALENCEMEMBER :em_expr (CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :em_rels (b) :em_is_const true :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({(RESTRICTINFO :clause {OEXPR :opno 98 :opfuncid 67 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({(VAR :varno 2 :varattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1) (CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 2) :required_rels (b 2) :incompatible_rels (b) :outer_rels (b) :left_rels (b 2) :right_rels (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1994 2095) :left_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 2 :varattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1) :em_rels (b 2) :em_is_const false :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}) :right_em (EQUIVALENCEMEMBER :em_expr (CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :em_rels (b) :em_is_const true :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({(VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1)})
WARNING: is_parallel_safe: OUTPUT qual = ({(VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1)}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({(RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({(VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1) (VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1) (VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1)}) :location -1) :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_rels (b 1 2) :required_rels (b 1 2) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b 2) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec 0.005 :outer_selec -1 :mergeopfamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1) :em_rels (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}) :right_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1) :em_rels (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}) :outer_is_left false :hashjoinoperator 96 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 96 :right_hashoperator 96})
WARNING: is_parallel_safe: OUTPUT qual = ({(RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({(VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1) (VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1) (VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1)}) :location -1) :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_rels (b 1 2) :required_rels (b 1 2) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b 2) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec 0.005 :outer_selec -1 :mergeopfamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1) :em_rels (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}) :right_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1) :em_rels (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}) :outer_is_left false :hashjoinoperator 96 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 96 :right_hashoperator 96}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1)})
WARNING: is_parallel_safe: OUTPUT qual = ({(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1)}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1)})
WARNING: is_parallel_safe: OUTPUT qual = ({(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1)}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {
  QUERY
  :commandType 1
  :querySource 0
  :canSetTag true
  :utilityStmt <>
  :resultRelation 0
  :hasAggs true
  :hasWindowFuncs false
  :hasTargetSRFs false
  :hasSubLinks true
  :hasDistinctOn false
  :hasRecursive false
  :hasModifyingCTE false
  :hasForUpdate false
  :hasRowSecurity true
  :hasGroupRE false
  :isReturn false
  :retlist <>
  :rttable (
    {
      RANGETBLENTY
      :alias <>
      :eref
      {
        ALIAS
        :aliasname partsupp
        :colnames ("ps_partkey" "ps_suppkty" "ps_availqty" "ps_supplycost" "p_s_comment")
      }
    }
  )
  :rtekind 0
  :relid 16556
  :inh false
  :relkind r
  :rellockmode 1
  :perminfoindex 1
  :tablesample <>
  :lateral false
  :inFromCl true
  :securityQuals (
    {
      OEXPR
      :opno 521
      :opfuncid 147
      :opresulttype 16
      :opretset false
      :opcollid 0
      :inputcollid 0
      :args (
        {
          VAR
          :varno 1
          :varattno 3
          :vartype 23
        }
      )
    }
  )
}
```

```

:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
(CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 100 0 0 0 0 0 0 ]
)
}
:location -1
}
{ALTERNATIVESUBPLAN
subplans (
{SUBPLAN
:subLinkType 0
:testexpr <>
:paramids <>
:plan_id 3
:plan_name SubPlan\ 3
:firstColType 2278
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam (i 1)
:args (
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
)
:startup_cost 0
:per_call_cost 485700.1124999999
}
{SUBPLAN
:subLinkType 2
:testexpr
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
{PARAM
:paramkind 1
:paramid 4
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
:paramids (i 4)
:plan_id 6
:plan_name SubPlan\ 6
:firstColType 23
:firstColTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam <>
:args <>
:startup_cost 485683.9999999999
:per_call_cost 0.0025
}
)
}
))
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname supplier
:colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
:rtekind 0
:relid 16541
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals ((
{OEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 6
:location -1
}
)
)
)
}

```

```

    }
    {CONST
    :consttype 701
    :consttypmod -1
    :constcollid 0
    :constlen 8
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 8 [ 0 0 0 0 0 0 0 0 ]
    }
  )
  :location -1
}
{ALTERNATIVESUBPLAN
:subplans (
  {SUBPLAN
  :subLinkType 0
  :testexpr <>
  :paramIds <>
  :plan_id 7
  :plan_name SubPlan\ 7
  :firstColType 2278
  :firstColTypmod -1
  :firstColCollation 0
  :useHashTable false
  :unknownEqFalse true
  :parallel_safe false
  :setParam <>
  :parParam (i 7)
  :args (
    {VAR
    :varno 2
    :varattno 4
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 4
    :location -1
    }
  )
  :startup_cost 0
  :per_call_cost 41
  }
  {SUBPLAN
  :subLinkType 2
  :testexpr
  {OPEXPR
  :opno 96
  :opfuncid 65
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
    {VAR
    :varno 2
    :varattno 4
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 4
    :location -1
    }
  )
  {PARAM
  :paramkind 1
  :paramid 8
  :paramtype 23
  :paramtypmod -1
  :paramcollid 0
  :location -1
  }
  )
  :location -1
  }
  :paramIds (i 8)
  :plan_id 8
  :plan_name SubPlan\ 8
  :firstColType 23
  :firstColTypmod -1
  :firstColCollation 0
  :useHashTable true
  :unknownEqFalse true
  :parallel_safe true
  :setParam <>
  :parParam <>
  :args <>
  :startup_cost 42.0124999999999996
  :per_call_cost 0.0025
  }
  )
}
))
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname nation
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtekind 0
:reldid 16536
:inh false
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname region
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:reldid 16531
:inh false
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:reldid 16556
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 11)
:insertedCols (b)
:updatedCols (b)
}
)

```

```

}
{RTEPERMISSIONINFO
:relid 16541
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16536
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16531
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
}
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
{RANGETBLREF
:rtindex 3
}
{RANGETBLREF
:rtindex 4
}
)
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{PARAM
:paramkind 1
:paramid 0
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location 812
}
{VAR
:verno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 824
}
)
:location 822
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:verno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 863
}
{VAR
:verno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 875
}
)
:location 873
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:verno 2
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 4
:location 914
}
{VAR
:verno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 928
}
}
}

```

```

    }
  }
  :location 926
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
:varno 3
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 3
:location 968
}
  {VAR
:varno 4
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location 982
}
)
:location 980
}
{OPEXPR
:opno 98
:opfuncid 67
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
  {VAR
:varno 4
:varattno 2
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location 1029
}
  {CONST
:consttype 25
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 1045
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
)
:location 1043
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
:expr
  {AGGREF
:aggfnoid 2136
:aggtype 701
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes (0 701)
:aggdirectargs <>
:args (
  {TARGETENTRY
:expr
  {VAR
:varno 1
:varattno 4
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 599
}
}
:resno 1
:resname <>
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar false
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggsplit 0
:aggno -1
:aggtransno -1
:location 595
}
:resno 1
:resname min
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>

```



```

WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 4 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -
1}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname n
:colnames <>
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtekind 0
:relid 16536
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname r
:colnames <>
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:relid 16531
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16536
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16531
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
)
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{PARAM
:paramkind 1
:paramid 9
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1

```

```

:varattnosyn 3
:location -1
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
}
:location -1
}
{OPEXPR
:opno 98
:opfuncid 67
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 2
:varattno 2
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location -1
}
{CONST
:consttype 25
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
)
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

```

WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 9 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_rels (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 9 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_rels (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})

WARNING: max_parallel_hazard_walker: Param paramid=9 in_safe_list=0 + reason=outer_reference

WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 9 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_rels (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 9 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_rels (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = false | reason = PROPARALLEL_RESTRICTED

WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {OPEXPR :opno 98 :opfuncid 67 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({VAR :varno 2 :varattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [40 0 0 0 69 85 82 79 80 69]}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 2) :required_rels (b 2) :incompatible_rels (b) :outer_rels (b) :left_rels (b 2) :right_rels (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1994 2095) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_rels (b 2) :em_is_const false :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [40 0 0 0 69 85 82 79 80 69]}) :em_rels (b) :em_is_const true :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE

WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} | parallel_safe = true | reason = PROPARALLEL_SAFE

WARNING: is_parallel_safe: INPUT qual = <>

WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE

WARNING: Log 2 : After pull up At starting of Path Generation:

DETAIL: {QUERY

```

:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRE false
:isReturn false
:costlist <>
:rtable (
{RANGETBLENTY
:alias
{ALIAS
:aliasname n

```

```

        :colnames <>
    }
}
:eref
{
    {ALIAS
        :aliasname n
        :colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
    }
}
:rtekind 0
:relid 16536
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTTRY
:alias
{
    {ALIAS
        :aliasname r
        :colnames <>
    }
}
}
:eref
{
    {ALIAS
        :aliasname r
        :colnames ("r_regionkey" "r_name" "r_comment")
    }
}
:rtekind 0
:relid 16531
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16536
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16531
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
}
)
:jointree
{
    {FROMEXPR
        :fromlist (
            {RANGETBLREF
                :rtindex 1
            }
            {RANGETBLREF
                :rtindex 2
            }
        )
    }
}
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
    {VAR
        :varno 1
        :varattno 3
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
    }
    :varnosyn 1
    :varattnosyn 3
    :location -1
    }
    {VAR
        :varno 2
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
    }
    :varnosyn 2
    :varattnosyn 1
    :location -1
    }
)
:location -1
}
{OPEXPR
:opno 98
:opfuncid 67
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
    {VAR
        :varno 2
        :varattno 2
        :vartype 25
        :vartypmod -1
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
    }
    :varnosyn 2
    :varattnosyn 2
    :location -1
    }
    {CONST
        :consttype 25
        :consttypmod -1
        :constcollid 100
        :constlen -1
        :constbyval false
        :constisnull false
        :location -1
        :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
    }
)
:location -1
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>

```

```

:targetList (
{TARGETENTRY
:expr
  (VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location -1
  )
  :resno 1
  :resname <>
  :ressortgroupref 0
  :resorigtbl 0
  :resorigcol 0
  :resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING:  is_parallel_safe: INPUT qual = <>
WARNING:  is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING:  is_parallel_safe: INPUT qual = (VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1)
(VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1)
WARNING:  is_parallel_safe: OUTPUT qual = (VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1)
(VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1) | parallel_safe = true | reason = PROP
ARALLEL_SAFE
WARNING:  is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {OEXPR :opno 98 :opfuncid 67 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ((VAR :varno 2 :varattno 2 :vartype 25 :
vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1) {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval
false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproo
f false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b 2) :outer_relds (b 2) :right_relds (b 2) :orclause <> :rinfo
_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1994 2095) :left_em {EQUIVALENCEMEMBER :em_expr (VAR :varno 2 :varattno 2 :vartype 25 :vartypmod
-1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1) :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 25 :em_jdomain {JOI
NDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 4
0 0 0 69 85 82 79 80 69 ]} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1
:right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0})
WARNING:  is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause {OEXPR :opno 98 :opfuncid 67 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ((VAR :varno 2 :varattno 2 :vartype 25
:vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1) {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyva
l false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b 2) :outer_relds (b 2) :right_relds (b 2) :orclause <> :rinfo
_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1994 2095) :left_em {EQUIVALENCEMEMBER :em_expr (VAR :varno 2 :varattno 2 :vartype 25 :vartypmo
d -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1) :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 25 :em_jdomain {JO
INDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [
40 0 0 69 85 82 79 80 69 ]} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -
1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING:  is_parallel_safe: INPUT qual = (VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1)
)
WARNING:  is_parallel_safe: OUTPUT qual = ((VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1
) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING:  is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 3 :vartype 23 :va
rtypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1) {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (
b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1) :location -1) :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false
:has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b 1 2) :incompatible_relds (b 2) :outer_relds (b 1) :right_relds (b 2) :orclause <> :rinfo_s
erial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec 0.005 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr (VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1
:varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1) :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMA
IN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr (VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattn
osyn 1 :location -1) :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :rig
ht_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 96 :right_hashoperator 96})
WARNING:  is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 3 :vartype 23 :v
artypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1) {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels
(b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1) :location -1) :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false
:has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b 1 2) :incompatible_relds (b 2) :outer_relds (b 1) :right_relds (b 2) :orclause <> :rinfo_s
erial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec 0.005 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr (VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1
:varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1) :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMA
IN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr (VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattn
osyn 1 :location -1) :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 96 :left_bucketsize -1 :ri
ght_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 96 :right_hashoperator 96}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING:  is_parallel_safe: INPUT qual = ((VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1)
)
WARNING:  is_parallel_safe: OUTPUT qual = ((VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING:  is_parallel_safe: INPUT qual = ((VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1)
)
WARNING:  is_parallel_safe: OUTPUT qual = ((VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING:  is_parallel_safe: INPUT qual = <>
WARNING:  is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING:  Log 2 : After pull up At starting of Path Generation:
DETAIL:
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTe false
:isReturn false
:ctelist <>
:ctable (
{RANGELENTTRY
:alias
{ALIAS
:aliasname n
:colnames <>
}
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
}
:rtekind 0
:reloid 16536
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}

```

```

{RANGETBLENTY
:alias
{alias
:aliasname r
:colnames <>
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:relid 16531
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16536
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16531
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtdindex 1
}
{RANGETBLREF
:rtdindex 2
}
)
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{PARAM
:paramkind 1
:paramid 12
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
)
:location -1
}
{OEXPR
:opno 98
:opfuncid 67
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 2
:varattno 2
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location -1
}
{CONST
:consttype 25
:consttypmod -1
:constcollid 100
}
)
}
}

```

```
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 96 :opfncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 12 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 12 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: max_parallel_hazard_walker: Param paramid=12 in_safe_list=0 → reason=outer_reference
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 96 :opfncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 12 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 12 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = false | reason = PROPARALLEL_RESTRICTED
WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 98 :opfncid 67 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({VAR :varno 2 :varattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b 2) :right_relds (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1994 2095) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :em_relds (b) :em_is_const true :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 98 :opfncid 67 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({VAR :varno 2 :varattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1994 2095) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :em_relds (b) :em_is_const true :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1}
)
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1}
) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTe false
:isReturn false
:tblist <>
:table (
{RANGEABLEENTRY
:alias
{ALIAS
:aliasname n
:colnames <>
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
}
:rtekind 0
:reloid 16536
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGEABLEENTRY
:alias
:aliasname r
:colnames <>
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
}
:rtekind 0
:reloid 16531
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:reloid 16536
}
```

```

:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16531
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
)
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
)
:location -1
}
{OEXPR
:opno 98
:opfuncid 67
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 2
:varattno 2
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location -1
}
{CONST
:consttype 25
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
)
:location -1
}
)
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1

```

```
:stmt_len -1
}

WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1)
{VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
{VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1}) | parallel_safe = true | reason = PROP
ARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({(RESTRICTINFO :clause {OEXPR :opno 98 :opfuncid 67 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({(VAR :varno 2 :varattno 2 :vartype 25 :
vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1) {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval
false :constisnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproo
f false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause <> :rinfo
_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1994 2095) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 25 :vartypmod
-1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 25 :em_jdomain {JOI
NDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [ 4
0 0 0 69 85 82 79 80 69 ]} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1
:right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({(RESTRICTINFO :clause {OEXPR :opno 98 :opfuncid 67 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({(VAR :varno 2 :varattno 2 :vartype 25
:vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1) {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyva
l false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause <> :rinf
o_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1994 2095) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 25 :vartypmod
d -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 25 :em_jdomain {JO
INDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [
40 0 0 69 85 82 79 80 69 ]} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -
1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({(VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1)
})
WARNING: is_parallel_safe: OUTPUT qual = ({(VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({(RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({(VAR :varno 1 :varattno 3 :vartype 23 :v
artypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1) {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (
b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1}) :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false
:has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b 1 2) :incompatible_relds (b) :outer_relds (b 1) :right_relds (b 2) :orclause <> :rinfo_s
erial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec 0.005 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1
:varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMA
IN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattn
osyn 1 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 96 :left_bucketsize -1 :rig
ht_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 96 :right_hashoperator 96})
WARNING: is_parallel_safe: OUTPUT qual = ({(RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({(VAR :varno 1 :varattno 3 :vartype 23 :v
artypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1) {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels
(b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false
:has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b 1 2) :incompatible_relds (b) :outer_relds (b 1) :right_relds (b 2) :orclause <> :rinfo
_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec 0.005 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1
:varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMA
IN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattn
osyn 1 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 96 :left_bucketsize -1 :rig
ht_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 96 :right_hashoperator 96}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1)
})
WARNING: is_parallel_safe: OUTPUT qual = ({(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1
}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL:
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE false
:isReturn false
:cteList <>
:table {
:RANGETABLEENTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_suppkty" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
:rtekind 0
:reldid 16541
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals ((
{OEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{CONST
:consttype 701
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 0 0 0 0 0 0 0 ]
}
})
:location -1
})
{ALTERNATIVESUBPLAN
:subplans (
{SUBPLAN
:sublinkType 0
:testexpr <>
:paramIds <>
:plan_id 12
:plan_name SubPlan\ 12
:firstColType 2278
:firstColTypmod -1
}
```

```

:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam (i 12)
:args (
  {VAR
    :varno 1
    :varattno 4
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 4
    :location -1
  }
)
:startup_cost 0
:per_call_cost 41
}
{SUBPLAN
:subLinkType 2
:testexpr
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 4
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 4
    :location -1
  }
  {PARAM
    :paramkind 1
    :paramid 13
    :paramtype 23
    :paramtypmod -1
    :paramcollid 0
    :location -1
  }
)
:location -1
}
:paramIds (i 13)
:plan_id 13
:plan_name SubPlan\ 13
:firstColType 23
:firstColTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 42.0124999999999996
:per_call_cost 0.0025
}
)
))
}
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :reloid 16541
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 0 13)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
{FROMEXPR
:fromlist (
  {RANGETBLREF
    :rtindex 1
  }
)
:quals (
  {OEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
        :location -1
      }
      {PARAM
        :paramkind 1
        :paramid 11
        :paramtype 23
        :paramtypmod -1
        :paramcollid 0
        :location -1
      }
    )
  }
  :location -1
)
{OEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 6
    :vartype 701
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
  }
)
}
}
}

```



```

        :aliasname r
        :colnames <>
    }
}
:eref
{
    (ALIAS
    :aliasname r
    :colnames ("r_regionkey" "r_name" "r_comment")
    )
}
:rtekind 0
:relid 16531
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16536
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16531
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
}
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
)
}
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{PARAM
:paramkind 1
:paramid 15
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
)
:location -1
}
{OEXPR
:opno 98
:opfuncid 67
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 2
:varattno 2
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location -1
}
{CONST
:consttype 25
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false

```

```

        :location -1
        :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
    }
}
:location -1
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

```

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 15 :paratype 23 :paramtypmod -1 :paramcollid 0 :location -1}} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 1 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 15 :paratype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})

WARNING: max_parallel_hazard_walker: Param paramid=15 in safe_list=> reason=outer_reference

WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 1 :location -1} {PARAM :paramkind 1 :paramid 15 :paratype 23 :paramtypmod -1 :paramcollid 0 :location -1}} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 1 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {PARAM :paramkind 1 :paramid 15 :paratype 23 :paramtypmod -1 :paramcollid 0 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = false | reason = PROPARALLEL_SAFE

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 98 :opfuncid 67 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args {(VAR :varno 2 :varrattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 2 :location -1} {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [40 0 0 0 69 85 82 79 80 69]}} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1994 2095) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varrattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :location -1 :constvalue 10 [40 0 0 0 69 85 82 79 80 69]}} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})

WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 98 :opfuncid 67 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args {(VAR :varno 2 :varrattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 2 :location -1} {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [40 0 0 0 69 85 82 79 80 69]}} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1994 2095) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varrattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 10 [40 0 0 0 69 85 82 79 80 69]}} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE

WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 2 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 1 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})

WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 2 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 1 :location -1} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})

WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE

WARNING: Log 2 : After pull up At starting of Path Generation:

```

DETAIL:
{QUERY
  :commandType 1
  :querySource 0
  :canSetTag true
  :utilityStmt <>
  :resultRelation 0
  :hasAggs false
  :hasWindowFuncs false
  :hasTargetSRFs false
  :hasSubLinks false
  :hasDistinctOn false
  :hasRecursive false
  :hasModifyingCTE false
  :hasForUpdate false
  :hasRowSecurity false
  :hasGroupRE false
  :isReturn false
  :cteList <>
  :rtable (
    {RANGELEFTENTRY
      :alias
      {ALIAS
        :aliasname n
        :colnames <>
      }
      :eref
      {ALIAS
        :aliasname n
        :colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
      }
      :rtekind 0
      :reldid 16536
      :inh false
      :relkind r
      :rellockmode 1
      :perminfoindex 1
      :tablesample <>
      :lateral false
      :inFromCl true
      :securityQuals <>
    }
    {RANGELEFTENTRY
      :alias
      {ALIAS
        :aliasname r
        :colnames <>
      }
      :eref
      {ALIAS
        :aliasname r
        :colnames ("r_regionkey" "r_name" "r_comment")
      }
      :rtekind 0
      :reldid 16531
      :inh false
      :relkind r
      :rellockmode 1
      :perminfoindex 2
      :tablesample <>
      :lateral false
      :inFromCl true
      :securityQuals <>
    }
  )
  :rtepermfos (
    {RTEPERMISSIONINFO
      :reldid 16536
      :inh true
      :requiredPerms 2
      :checkAsUser 0
    }
  )
}

```

```

:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16531
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
)
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
)
:location -1
}
{OEXPR
:opno 98
:opfuncid 67
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 2
:varattno 2
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location -1
}
{CONST
:consttype 25
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
)
:location -1
}
)
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

```

```
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1})
(VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1})
(VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1}) | parallel_safe = true | reason = PROP
ARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OPEXPR :opno 98 :opfuncid 67 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({VAR :varno 2 :varattno 2 :vartype 25 :
vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1}) (CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval
false :constnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproo
f false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause <> :rinfo
_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1994 2095) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 25 :vartypmod
-1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 25 :em_jdomain {JOI
NDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constnull false :location -1 :constvalue 10 [ 4
0 0 0 69 85 82 79 80 69 ]} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1
:right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OPEXPR :opno 98 :opfuncid 67 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({VAR :varno 2 :varattno 2 :vartype 25
:vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1}) (CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyva
l false :constnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakpro
of false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b) :orclause <> :rinfo
_o_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies (o 1994 2095) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 2 :vartype 25 :vartypmod
-1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 2 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 25 :em_jdomain {JOI
NDOMAIN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constnull false :location -1 :constvalue 10 [
40 0 0 0 69 85 82 79 80 69 ]} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -
1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1})
(VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1})
(VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1})
WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 3 :vartype 23 :v
artypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1}) (CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyva
l false :constnull false :location -1 :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]}) :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false
:has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b 1 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b 2) :orclause <> :rinfo_s
erial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec 0.005 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1
:varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMA
IN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constnull false :location -1 :constvalue 10 [ 40
0 0 0 69 85 82 79 80 69 ]} :em_relds (b) :em_is_const true :em_is_child false :em_datatype 25 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :rig
ht_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 96 :right_hasheqoperator 96})
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 3 :vartype 23 :v
artypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1}) (CONST :consttype 25 :consttypmod -1 :constcollid 0 :varnullingrels
(b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1}) :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false
:has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 2) :required_relds (b 1 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b 2) :orclause <> :rinfo_
serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec 0.005 :outer_selec -1 :mergeopfamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1
:varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMA
IN :jd_relds (b 1 2)}} :right_em {EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varatt
nosyn 1 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2)}} :outer_is_left false :hashjoinoperator 96 :left_bucketsize -1 :ri
ght_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 96 :right_hasheqoperator 96}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1})
(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1})
(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1})
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1})
(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1})
(VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1})
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL:
    QUERY
    :commandType 1
    :querySource 0
    :canSetTag true
    :utilityStmt <>
    :resultRelation 0
    :hasAggs false
    :hasWindowFuncs false
    :hasTargetSRFs false
    :hasSublinks true
    :hasDistinctOn false
    :hasRecursive false
    :hasModifyingCTE false
    :hasForUpdate false
    :hasRowSecurity true
    :hasGroupRTF false
    :isReturn false
    :testlist <>
    :rtable (
      :RANGELEFTENTRY
      :alias
      {ALIAS
        :aliasname s
        :colnames <>
      }
      :eref
      {ALIAS
        :aliasname s
        :colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "
          s_acctbal" "s_comment")
      }
      :rtekind 0
      :reld 16541
      :inh false
      :relkind r
      :rellockmode 1
      :perminfoindex 1
      :tablesample <>
      :lateral false
      :inFromCl true
      :securityQuals ((
        {OPEXPR
          :opno 674
          :opfuncid 297
          :opresulttype 16
          :opretset false
          :opcollid 0
          :inputcollid 0
          :args (
            (VAR
              :varno 1
              :varattno 6
              :vartype 701
              :vartypmod -1
              :varcollid 0
              :varnullingrels (b)
              :varlevelsup 0
              :varreturningtype 0
              :varnosyn 1
              :varattnosyn 6
              :location -1
            )
            (CONST
              :consttype 701
              :consttypmod -1
              :constcollid 0
              :constlen 8
              :constbyval true
              :constnull false
              :location -1
              :constvalue 8 [ 0 0 0 0 0 0 0 0 ]
            )
          )
        )
      :location -1
    )
    :ALTERNATIVESUBPLAN
    :subplans (
      :SUBPLAN
      :subLinkType 0
      :testexpr <>
      :paramIds <>
      :plan_id 15
      :plan_name SubPlan\ 15
      :firstColType 2278
      :firstColTypmod -1
      :firstColCollation 0
      :useHashTable false
      :unknownEqFalse true
    )
  )
}
```

```

:parallel_safe false
:setParam <>
:parParam (i 15)
:args (
  {VAR
    :varno 1
    :varattno 4
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 4
    :location -1
  }
)
:startup_cost 0
:per_call_cost 41
}
{SUBPLAN
:sublinkType 2
:testexpr
  {OEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 1
        :varattno 4
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 4
        :location -1
      }
      {PARAM
        :paramkind 1
        :paramid 16
        :paramtype 23
        :paramtypmod -1
        :paramcollid 0
        :location -1
      }
    )
    :location -1
  }
:paramids (i 16)
:plan_id 16
:plan_name SubPlan\ 16
:firstColType 23
:firstColTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 42.012499999999996
:per_call_cost 0.0025
}
)
}
))
}
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :reld 16541
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 13)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
  {FROMEXPR
    :fromlist (
      {RANGETBLREF
        :rtindex 1
      }
    )
  }
:quals (
  {OEXPR
    :opno 674
    :opfuncid 297
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 1
        :varattno 6
        :vartype 701
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 6
        :location -1
      }
      {CONST
        :consttype 701
        :consttypmod -1
        :constcollid 0
        :constlen 8
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 8 [ 0 0 0 0 0 0 0 0 ]
      }
    )
    :location -1
  }
)
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
      {VAR
        :varno 1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
      }
    }
  )

```

```

        :location -1
      }
      :resno 1
    <:resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
}

:override 0
:onConflict <>
:ReturningOldAlias <>
:ReturningNewAlias <>
:ReturningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
:windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {OEXPR :opno 674 :opfuncid 297 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1) (CONST :consttype 701 :consttypmod -1 :constcollid 0 :constlen 8 :constbyval true :constisnull false :location -1 :constvalue 8 [ 0 0 0 0 0 0 0 0 ]}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_mvcfreq 0 :right_mvcfreq 0) {RESTRICTINFO :clause {ALTERNATIVESUBPLAN :subplans ((SUBPLAN :sublinkType 0 :testexpr <> :paramids <> :plan_id 15 :plan_name SubPlan 15 :firstColType 2278 :firstColTypmod -1 :firstColCollation 0 :useHashTable false :unknownEqFalse true :parallel_safe false :setParam <> :parParam (i 15) :args ((VAR :varno 1 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1) :startup_cost 0 :per_call_cost 41) {SUBPLAN :sublinkType 2 :testexpr {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1) (PARAM :paramkind 1 :paramid 16 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1) :location -1) :paramids (i 16) :plan_id 16 :plan_name SubPlan 16 :firstColType 23 :firstColTypmod -1 :firstColCollation 0 :useHashTable true :unknownEqFalse true :parallel_safe true :setParam <> :parParam <> :args <> :startup_cost 42.812499999999996 :per_call_cost 0.0025)) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b) :right_rels (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_mvcfreq 0 :right_mvcfreq 0) :right_hashjoinoperator 0} :right_hashjoinoperator 0})
WARNING: max_parallel_hazard_walker: SubPlan plan_id=15 parallel_safe=0 → reason=correlated_subplan
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause {OEXPR :opno 674 :opfuncid 297 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1) (CONST :consttype 701 :consttypmod -1 :constcollid 0 :constlen 8 :constbyval true :constisnull false :location -1 :constvalue 8 [ 0 0 0 0 0 0 0 0 ]}) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b 1) :right_rels (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_mvcfreq 0 :right_mvcfreq 0) {SUBPLAN :sublinkType 0 :testexpr <> :paramids <> :plan_id 15 :plan_name SubPlan 15 :firstColType 2278 :firstColTypmod -1 :firstColCollation 0 :useHashTable false :unknownEqFalse true :parallel_safe false :setParam <> :parParam (i 15) :args ((VAR :varno 1 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1) :startup_cost 0 :per_call_cost 41) {SUBPLAN :sublinkType 2 :testexpr {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 4 :location -1) (PARAM :paramkind 1 :paramid 16 :paramtype 23 :paramtypmod -1 :paramcollid 0 :location -1) :location -1) :paramids (i 16) :plan_id 16 :plan_name SubPlan 16 :firstColType 23 :firstColTypmod -1 :firstColCollation 0 :useHashTable true :unknownEqFalse true :parallel_safe true :setParam <> :parParam <> :args <> :startup_cost 42.812499999999996 :per_call_cost 0.0025)) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b) :left_rels (b) :right_rels (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_mvcfreq 0 :right_mvcfreq 0) :right_hashjoinoperator 0} :right_hashjoinoperator 0} | parallel_safe = false | reason = PROPAGALLED_RESTRICTED
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1)
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1)
) | parallel_safe = true | reason = PROPAGALLED_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL:
  QUERY
  commandType 1
  querySource 0
  canSetTag true
  utilityStmt <>
  resultRelation 0
  hasAggs false
  hasWindowFuncs false
  hasTargetSRFs false
  hasSublinks true
  hasDistinctOn false
  hasRecursive false
  hasModifyingCTE false
  hasForUpdate false
  hasRowSecurity true
  hasGroupRTE false
  isReturn false
  retlist <>
  rtable (
    {RANGETABLEENTRY
      alias <>
      eref
      {ALIAS
        aliasname part
        colnames ("p_partkey" "p_name" "p_mfg" "p_brand" "p_type" "p_size"
          "p_container" "p_retailprice" "p_comment")
      }
      rtekind 0
      relid 16551
      inh false
      relkind r
      rellockmode 1
      perminfoindex 1
      tablesample <>
      lateral false
      infFromCl true
      securityQuals <>
    }
    {RANGETABLEENTRY
      alias <>
      eref
      {ALIAS
        aliasname supplier
        colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "
          s_acctbal" "s_comment")
      }
      rtekind 0
      relid 16541
      inh false
      relkind r
      rellockmode 1
      perminfoindex 2
      tablesample <>
      lateral false
      infFromCl true
      securityQuals ((
        {OEXPR
          opno 674
          opfuncid 297
          opresulttype 16
          opretset false
          opcollid 0
          inputcollid 0
          args (
            {VAR
              varno 2
              varattno 6
              vartype 701
              vartypmod -1
            }
          )
        }
      )
    )
  )

```

```

:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 6
:location -1
}
(CONST
:consttype 781
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 0 0 0 0 0 0 0 ]
)
:location -1
}
{ALTERNATIVESUBPLAN
:subplans (
(SUBPLAN
:sublinkType 0
:testexpr <>
:paramids <>
:plan_id 10
:plan_name SubPlan\ 10
:firstColType 2278
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam (i 9)
:args (
{VAR
:varno 2
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 4
:location -1
}
)
:startup_cost 0
:per_call_cost 41
}
(SUBPLAN
:sublinkType 2
:testexpr
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 4
:location -1
}
)
{PARAM
:paramkind 1
:paramid 10
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
:paramids (i 10)
:plan_id 11
:plan_name SubPlan\ 11
:firstColType 23
:firstColTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 42.812499999999996
:per_call_cost 0.0025
}
)
}
))
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname partsupp
:colnames ("ps_partkey" "ps_suppkey" "ps_availqty" "ps_supplycost" "p
s_comment")
}
:rtkind 0
:relid 16556
:inh false
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals ((
{OPEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 3
:location -1
}
)

```

```

(CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 100 0 0 0 0 0 0 ]
)
:location -1
}
{ALTERNATIVESUBPLAN
:subplans (
{SUBPLAN
:sublinkType 0
:testexpr <>
:paramids <>
:plan_id 14
:plan_name SubPlan\ 14
:firstColType 2278
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam (i 11)
:args (
{VAR
:varno 3
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 2
:location -1
}
)
:startup_cost 0
:per_call_cost 485700.1124999999
}
{SUBPLAN
:sublinkType 2
:testexpr
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 2
:location -1
}
{PARAM
:paramkind 1
:paramid 14
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
:paramids (i 14)
:plan_id 17
:plan_name SubPlan\ 17
:firstColType 23
:firstColTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam <>
:args <>
:startup_cost 485683.9999999999
:per_call_cost 0.0025
)
}
))
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname nation
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtekind 0
:relid 16536
:inh false
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname region
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:relid 16531
:inh false
:relkind r
:rellockmode 1
:perminfoindex 5
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rtoperinfos (
{RTEPERMISSIONINFO
:relid 16551
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10 12 13)
:insertedCols (b)
:updatedCols (b)
}
)

```

```

{RTEPERMISSIONINFO
:relid 16541
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16556
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16536
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16531
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
}
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
{RANGETBLREF
:rtindex 3
}
{RANGETBLREF
:rtindex 4
}
{RANGETBLREF
:rtindex 5
}
)
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 283
}
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 295
}
)
:location 293
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 318
}
{VAR
:varno 3
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 2
:location 330
}
)
:location 328
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0

```

```

:varnosyn 1
:varattnosyn 6
:location 353
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 362
:constvalue 4 [ 15 0 0 0 0 0 0 ]
}
)
:location 360
}
{OEXPR
:opno 1209
:opfuncid 850
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 1
:varattno 5
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 377
}
{CONST
:consttype 25
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 389
:constvalue 10 [ 40 0 0 0 37 66 82 65 83 83 ]
}
)
:location 384
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 4
:location 410
}
{VAR
:varno 4
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location 424
}
)
:location 422
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 3
:location 448
}
{VAR
:varno 5
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location 462
}
)
:location 460
}
{OEXPR
:opno 98
:opfuncid 67
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 5
:varattno 2
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 2
:location 493
}
{CONST
:consttype 25

```

```

:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 509
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
}
location 507
}
{OEXPR
:opno 670
:opfuncid 293
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 4
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 4
:location 530
}
{SUBPLAN
:sublinkType 4
:testexpr <>
:paramsids <>
:plan_id 9
:plan_name SubPlan\ 9
:firstColType 701
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse false
:parallel_safe false
:setParam <>
:parParam (i 0)
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 812
}
}
:startup_cost 0
:per_call_cost 426611327037.82996
}
}
location 544
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 6
:location 51
}
:resno 1
:resname s_acctbal
:ressortgroupref 1
:resorigtbl 16541
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 2
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location 70
}
:resno 2
:resname s_name
:ressortgroupref 3
:resorigtbl 16541
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 2
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location 86
}
:resno 3
:resname n_name
:ressortgroupref 2
:resorigtbl 16536
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)

```

```

:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 102
}
:resno 4
:resname p_partkey
:ressortgroupref 4
:resorigtbl 16551
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 121
}
:resno 5
:resname p_mfgr
:ressortgroupref 0
:resorigtbl 16551
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 3
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 3
:location 137
}
:resno 6
:resname s_address
:ressortgroupref 0
:resorigtbl 16541
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 5
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 5
:location 156
}
:resno 7
:resname s_phone
:ressortgroupref 0
:resorigtbl 16541
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 7
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 7
:location 173
}
:resno 8
:resname s_comment
:ressortgroupref 0
:resorigtbl 16541
:resorigcol 7
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
{SORTGROUPCLAUSE
:tltSortGroupRef 1
:eqop 670
:sortop 674
:reverse_sort true
:nulls_first true
:hashable true
}
{SORTGROUPCLAUSE
:tltSortGroupRef 2
:eqop 98
:sortop 664
:reverse_sort false
:nulls_first false
:hashable true
}
{SORTGROUPCLAUSE
:tltSortGroupRef 3
:eqop 98
:sortop 664
:reverse_sort false
:nulls_first false
:hashable true
}
{SORTGROUPCLAUSE
:tltSortGroupRef 4
:eqop 96
:sortop 97
:reverse_sort false
:nulls_first false
:hashable true
}
)
:limitOffset <>

```



```
has_volatile 2 :security_level 1 :num_base_rels 2 :clause_relds (b 4 5) :required_relds (b 4 5) :incompatible_relds (b) :outer_relds (b) :left_relds (b 4) :right_relds (b 5) :orclause <> :rinfo_se
rial 16 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select 0.005 :outer_select -1 :mergeopfamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 4 :varrattno 3 :vartype 23 :vartypmod -1
:varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varrattnosyn 3 :location -1) :em_relds (b 4) :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMA1
W :jd_relds (b 1 2 3 4 5)}) :right_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 5 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :va
rattnosyn 1 :location -1) :em_relds (b 5) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2 3 4 5)}) :outer_is_left false :hashjoinoperator 96 :left_bucket
size -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 96 :right_hashoperator 96)) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICINFO :clause (OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 4 :varrattno 3 :vartype 23 :v
artypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varrattnosyn 3 :location -1) (VAR :varno 5 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels
(b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varrattnosyn 1 :location -1) :location -1) :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof true
:has_volatile 2 :security_level 1 :num_base_rels 2 :clause_relds (b 4 5) :required_relds (b 4 5) :incompatible_relds (b) :outer_relds (b 4) :right_relds (b 5) :orclause <> :rinfo_s
erial 16 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_select 0.005 :outer_select -1 :mergeopfamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 4 :varrattno 3 :vartype 23 :vartypmod -1
:varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varrattnosyn 3 :location -1) :em_relds (b 4) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMA
IN :jd_relds (b 1 2 3 4 5)}) :right_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 5 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :v
arattnosyn 1 :location -1) :em_relds (b 5) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 1 2 3 4 5)}) :outer_is_left false :hashjoinoperator 96 :left_bucke
tsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 96 :right_hashoperator 96)) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 4 :varrattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varrattnosyn 2 :location -
1) (VAR :varno 4 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varrattnosyn 1 :location -1) :location -1)
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 4 :varrattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varrattnosyn 2 :location
-1) (VAR :varno 4 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varrattnosyn 1 :location -1) | parallel_safe = true | reason = PR
OPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 2 :varrattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 6 :location -1
) (VAR :varno 2 :varrattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 2 :location -1) (VAR :varno 4 :varrattno 2 :vartype 25
:vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varrattnosyn 2 :location -1) (VAR :varno 1 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullin
grels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 1 :location -1) (VAR :varno 1 :varrattno 3 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturning
pe 0 :varnosyn 1 :varrattnosyn 3 :location -1) (VAR :varno 2 :varrattno 3 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 3 :locati
on -1) (VAR :varno 2 :varrattno 5 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 5 :location -1) (VAR :varno 2 :varrattno 7 :vartyp
pe 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 7 :location -1)
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 2 :varrattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 6 :location -
1) (VAR :varno 2 :varrattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 2 :location -1) (VAR :varno 4 :varrattno 2 :vartype 25
:vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varrattnosyn 2 :location -1) (VAR :varno 1 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullin
grels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 1 :location -1) (VAR :varno 1 :varrattno 3 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturning
type 0 :varnosyn 1 :varrattnosyn 3 :location -1) (VAR :varno 2 :varrattno 3 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 3 :locati
on -1) (VAR :varno 2 :varrattno 5 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 5 :location -1) (VAR :varno 2 :varrattno 7 :vartyp
pe 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 7 :location -1)
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 2 :varrattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 6 :location -1
) (VAR :varno 2 :varrattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 2 :location -1) (VAR :varno 4 :varrattno 2 :vartype 25
:vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varrattnosyn 2 :location -1) (VAR :varno 1 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullin
grels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 1 :location -1) (VAR :varno 1 :varrattno 3 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturning
type 0 :varnosyn 1 :varrattnosyn 3 :location -1) (VAR :varno 2 :varrattno 3 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 3 :locati
on -1) (VAR :varno 2 :varrattno 5 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 5 :location -1) (VAR :varno 2 :varrattno 7 :vartyp
pe 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 7 :location -1)
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 2 :varrattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 6 :location -
1) (VAR :varno 2 :varrattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 2 :location -1) (VAR :varno 4 :varrattno 2 :vartype 25
:vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varrattnosyn 2 :location -1) (VAR :varno 1 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullin
grels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 1 :location -1) (VAR :varno 1 :varrattno 3 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturning
type 0 :varnosyn 1 :varrattnosyn 3 :location -1) (VAR :varno 2 :varrattno 3 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 3 :locati
on -1) (VAR :varno 2 :varrattno 5 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 5 :location -1) (VAR :varno 2 :varrattno 7 :vartyp
pe 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 7 :location -1) | parallel_safe = true | reason = PROPARALLEL_SAFE
QUERY PLAN
-----
Limit (cost=187282375428557.16..187282375428557.16 rows=1 width=182) (actual time=680909.023..680912.032 rows=100.00 loops=1)
  Output: supplier.s_acctbal, supplier.s_name, nation.n_name, part.p_partkey, part.p_mfgr, supplier.s_address, supplier.s_phone, supplier.s_comment
  Buffers: shared hit=448303 read=92488500 dirtied=415 written=61, temp read=24481 written=24481
  -> Sort (cost=187282375428557.16..187282375428557.16 rows=1 width=182) (actual time=680909.021..680911.764 rows=100.00 loops=1)
    Output: supplier.s_acctbal, supplier.s_name, nation.n_name, part.p_partkey, part.p_mfgr, supplier.s_address, supplier.s_phone, supplier.s_comment
    Sort Key: supplier.s_acctbal DESC, nation.n_name, supplier.s_name, part.p_partkey
    Sort Method: top-N heapsort  Memory: 69kB
    Buffers: shared hit=448303 read=92488500 dirtied=415 written=61, temp read=24481 written=24481
    -> Hash Join (cost=426614178491.79..187282375428557.16 rows=1 width=182) (actual time=170752.515..680899.550 rows=2440.00 loops=1)
      Output: supplier.s_acctbal, supplier.s_name, nation.n_name, part.p_partkey, part.p_mfgr, supplier.s_address, supplier.s_phone, supplier.s_comment
      Hash Cond: (((SubPlan 9) = partsupp.ps_supplycost) AND (part.p_partkey = partsupp.ps_partkey))
      Buffers: shared hit=448294 read=92488500 dirtied=415 written=61, temp read=24481 written=24481
      -> Seq Scan on public.part (cost=0.00..7489.28 rows=876 width=19) (actual time=1.004..76.581 rows=820.00 loops=1)
        Output: part.p_partkey, part.p_name, part.p_mfgr, part.p_brand, part.p_type, part.p_size, part.p_container, part.p_retailprice, part.p_comment
        Filter: ((part.p_size = 15) AND (part.p_type ~ 'BKRASS')::text)
        Rows Removed by Filter: 211980
        Buffers: shared read=4216 dirtied=383 written=61
      -> Hash (cost=426614178251.01..426614178251.01 rows=16052 width=175) (actual time=166020.202..166022.154 rows=748884.00 loops=1)
        Output: supplier.s_acctbal, supplier.s_name, supplier.s_address, supplier.s_phone, supplier.s_comment, partsupp.ps_partkey, partsupp.ps_supplycost, nation.n_name
        Buckets: 65936 (originally 16384) Batches: 32 (originally 11) Memory Usage: 7681kB
        Buffers: shared hit=262 read=20044 dirtied=32, temp written=15654
        -> Nested Loop (cost=0.00..426614178251.01 rows=16052 width=175) (actual time=32.663..165082.193 rows=748884.00 loops=1)
          Output: supplier.s_acctbal, supplier.s_name, supplier.s_address, supplier.s_phone, supplier.s_comment, partsupp.ps_partkey, partsupp.ps_supplycost, nation.n_name
          Join Filter: (supplier.s_supplykey = partsupp.ps_supplykey)
          Rows Removed by Join Filter: 1258789132
          Buffers: shared hit=262 read=20044 dirtied=32
          -> Seq Scan on public.partsupp (cost=0.00..426610839976.75 rows=435124 width=16) (actual time=24.646..967.004 rows=157916.00 loops=1)
            Output: partsupp.ps_partkey, partsupp.ps_supplykey, partsupp.ps_availqty, partsupp.ps_supplycost, partsupp.ps_comment
            Filter: ((partsupp.ps_availqty > 100) AND (ANY (partsupp.ps_supplykey = (hashed SubPlan 17).col1)))
            Rows Removed by Filter: 722084
            Buffers: shared read=20044 dirtied=32
            SubPlan 17
            -> Seq Scan on public.supplier_s_1 (cost=0.00..485670.52 rows=5390 width=4) (actual time=3.846..24.102 rows=1994.00 loops=1)
              Output: s_1.s_supplykey
              Filter: ((s_1.s_acctbal > '0')::double precision) AND (s_1.s_acctbal > '0')::double precision) AND (ANY (s_1.s_nationkey = (hashed SubPlan 16).col1)))
              Rows Removed by Filter: 9006
              Buffers: shared read=260 dirtied=32
              SubPlan 16
              -> Hash Join (cost=20.68..41.97 rows=16 width=4) (actual time=1.927..1.950 rows=20.00 loops=1)
                Output: n_3.n_nationkey
                Hash Cond: (n_3.n_regionkey = r_3.r_regionkey)
                Buffers: shared read=2 dirtied=2
                -> Seq Scan on public.nation_n_3 (cost=0.00..18.10 rows=810 width=8) (actual time=1.102..1.111 rows=50.00 loops=1)
                  Output: n_3.n_nationkey, n_3.n_name, n_3.n_regionkey, n_3.n_comment
                  Buffers: shared read=1 dirtied=1
                -> Hash (cost=20.62..20.62 rows=4 width=4) (actual time=0.754..0.757 rows=2.00 loops=1)
                  Output: r_3.r_regionkey
                  Buckets: 1024 Batches: 1 Memory Usage: 9KB
                  Buffers: shared read=1 dirtied=1
                  -> Seq Scan on public.region_r_3 (cost=0.00..20.62 rows=4 width=4) (actual time=0.723..0.725 rows=2.00 loops=1)
                    Output: r_3.r_regionkey
                    Filter: (r_3.r_name = 'EUROPE')::text)
                    Rows Removed by Filter: 8
                    Buffers: shared read=1 dirtied=1
      -> Materialize (cost=0.00..487037.53 rows=437 width=167) (actual time=0.000..0.297 rows=7976.00 loops=157916)
        Output: supplier.s_acctbal, supplier.s_name, supplier.s_address, supplier.s_phone, supplier.s_comment, supplier.s_supplykey, nation.n_name
        Storage: Memory Maximum Storage: 1491kB
        Buffers: shared hit=262
        -> Nested Loop (cost=0.00..487035.35 rows=437 width=167) (actual time=0.148..13.683 rows=7976.00 loops=1)
          Join Filter: (supplier.s_nationkey = nation.n_nationkey)
          Rows Removed by Join Filter: 31904
          Buffers: shared hit=262
          -> Nested Loop (cost=0.00..87.34 rows=16 width=36) (actual time=0.044..0.072 rows=20.00 loops=1)
            Output: nation.n_name, nation.n_nationkey
            Join Filter: (nation.n_regionkey = region.r_regionkey)
            Rows Removed by Join Filter: 00
            Buffers: shared hit=2
            -> Seq Scan on public.nation (cost=0.00..18.10 rows=810 width=40) (actual time=0.012..0.016 rows=50.00 loops=1)
              Output: nation.n_nationkey, nation.n_name, nation.n_regionkey, nation.n_comment
              Buffers: shared hit=1
            -> Materialize (cost=0.00..20.64 rows=4 width=4) (actual time=0.001..0.001 rows=2.00 loops=50)
              Output: region.r_regionkey
              Storage: Memory Maximum Storage: 17kB
              Buffers: shared hit=1
              -> Seq Scan on public.region (cost=0.00..20.62 rows=4 width=4) (actual time=0.008..0.009 rows=2.00 loops=1)
                Output: region.r_regionkey
                Filter: (region.r_name = 'EUROPE')::text)
                Rows Removed by Filter: 8
                Buffers: shared hit=1
      -> Materialize (cost=0.00..485667.89 rows=5390 width=139) (actual time=0.004..0.356 rows=1994.00 loops=20)
        Output: supplier.s_acctbal, supplier.s_name, supplier.s_address, supplier.s_phone, supplier.s_comment, supplier.s_supplykey, supplier.s_nationkey
        Storage: Memory Maximum Storage: 367kB
        Buffers: shared hit=260
        -> Seq Scan on public.supplier (cost=0.00..485640.94 rows=5390 width=139) (actual time=0.068..3.630 rows=1994.00 loops=1)
          Output: supplier.s_acctbal, supplier.s_name, supplier.s_address, supplier.s_phone, supplier.s_comment, supplier.s_supplykey, supplier.s_nationkey
          Filter: (((supplier.s_acctbal > '0')::double precision) AND (ANY (supplier.s_nationkey = (hashed SubPlan 11).col1)))
          Rows Removed by Filter: 9006
          Buffers: shared hit=260
          SubPlan 11
          -> Hash Join (cost=20.68..41.97 rows=16 width=4) (actual time=0.033..0.732 rows=20.00 loops=1)
            Output: n_2.n_nationkey
            Hash Cond: (n_2.n_regionkey = r_2.r_regionkey)
            Buffers: shared hit=2
            -> Seq Scan on public.nation_n_2 (cost=0.00..18.10 rows=810 width=8) (actual time=0.005..0.344 rows=50.00 loops=1)
              Output: n_2.n_nationkey, n_2.n_name, n_2.n_regionkey, n_2.n_comment
              Buffers: shared hit=1
            -> Hash (cost=20.62..20.62 rows=4 width=4) (actual time=0.014..0.015 rows=2.00 loops=1)
```

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Output: r_2.r_regionkey
Buckets: 1024 Batches: 1 Memory Usage: 9kB
Buffers: shared hit=1
-> Seq Scan on public.region r_2 (cost=0.00..20.62 rows=4 width=4) (actual time=0.003..0.004 rows=2.00 loops=1)
    Output: r_2.r_regionkey
    Filter: (r_2.r_name = 'EUROPE'::text)
    Rows Removed by Filter: 8
    Buffers: shared hit=1

SubPlan 9
-> Aggregate (cost=426611327037.82..426611327037.83 rows=1 width=8) (actual time=109.507..109.507 rows=1.00 loops=4696)
    Output: min(partsupp_1.ps_supplycost)
    Buffers: shared hit=448032 read=92464240
    -> Nested Loop (cost=0.00..426611327037.82 rows=1 width=8) (actual time=70.019..109.502 rows=5.18 loops=4696)
        Output: partsupp_1.ps_supplycost
        Join Filter: (nation_1.n_regionkey = region_1.r_regionkey)
        Buffers: shared hit=448032 read=92464240
        -> Nested Loop (cost=0.00..426611327037.82 rows=4 width=12) (actual time=70.018..109.500 rows=2.59 loops=4696)
            Output: partsupp_1.ps_supplycost, nation_1.n_regionkey
            Join Filter: (supplier_1.s_nationkey = nation_1.n_nationkey)
            Rows Removed by Join Filter: 62
            Buffers: shared hit=448031 read=92464240
            -> Nested Loop (cost=0.00..426611326988.72 rows=1 width=12) (actual time=70.010..109.481 rows=1.30 loops=4696)
                Output: partsupp_1.ps_supplycost, supplier_1.s_nationkey
                Join Filter: (partsupp_1.ps_suppkey = supplier_1.s_suppkey)
                Rows Removed by Join Filter: 1983
                Buffers: shared hit=441944 read=92464240
                -> Seq Scan on public.partsupp partsupp_1 (cost=0.00..426610841172.60 rows=2 width=12) (actual time=69.953..109.275 rows=1.00 loops=4696)
                    Output: partsupp_1.ps_partkey, partsupp_1.ps_suppkey, partsupp_1.ps_availqty, partsupp_1.ps_supplycost, partsupp_1.ps_comment
                    Filter: ((partsupp_1.ps_availqty > 100) AND (part.p_partkey = partsupp_1.ps_partkey) AND (ANY (partsupp_1.ps_suppkey = (hashed SubPlan 6).col1))))
                    Rows Removed by Filter: 879999
                    Buffers: shared hit=441684 read=92464240
                    SubPlan 6
                    -> Seq Scan on public.supplier s (cost=0.00..485670.52 rows=5390 width=4) (actual time=0.821..5.084 rows=1994.00 loops=1)
                        Output: s.s_suppkey
                        Filter: ((s.s_acctbal > '0'::double precision) AND (s.s_acctbal > '0'::double precision) AND (ANY (s.s_nationkey = (hashed SubPlan 5).col1))))
                        Rows Removed by Filter: 9006
                        Buffers: shared hit=260
                        SubPlan 5
                        -> Hash Join (cost=20.68..41.97 rows=16 width=4) (actual time=0.326..0.350 rows=20.00 loops=1)
                            Output: n.n_nationkey
                            Hash Cond: (n.n_regionkey = r.r_regionkey)
                            Buffers: shared hit=2
                            -> Seq Scan on public.nation n (cost=0.00..18.10 rows=810 width=8) (actual time=0.010..0.016 rows=50.00 loops=1)
                                Output: n.n_nationkey, n.n_name, n.n_regionkey, n.n_comment
                                Buffers: shared hit=1
                            -> Hash (cost=20.62..20.62 rows=4 width=4) (actual time=0.300..0.301 rows=2.00 loops=1)
                                Output: r.r_regionkey
                                Buckets: 1024 Batches: 1 Memory Usage: 9kB
                                Buffers: shared hit=1
                                -> Seq Scan on public.region r (cost=0.00..20.62 rows=4 width=4) (actual time=0.298..0.291 rows=2.00 loops=1)
                                    Output: r.r_regionkey
                                    Filter: (r.r_name = 'EUROPE'::text)
                                    Rows Removed by Filter: 8
                                    Buffers: shared hit=1
                -> Materialize (cost=0.00..485667.89 rows=5390 width=8) (actual time=0.001..0.081 rows=1994.00 loops=4674)
                    Output: supplier_1.s_suppkey, supplier_1.s_nationkey
                    Storage: Memory Maximum Storage: 110kB
                    Buffers: shared hit=260
                    -> Seq Scan on public.supplier supplier_1 (cost=0.00..485640.94 rows=5390 width=8) (actual time=0.199..2.844 rows=1994.00 loops=1)
                        Output: supplier_1.s_suppkey, supplier_1.s_nationkey
                        Filter: ((supplier_1.s_acctbal > '0'::double precision) AND (ANY (supplier_1.s_nationkey = (hashed SubPlan 8).col1))))
                        Rows Removed by Filter: 9006
                        Buffers: shared hit=260
                        SubPlan 8
                        -> Hash Join (cost=20.68..41.97 rows=16 width=4) (actual time=0.096..0.512 rows=20.00 loops=1)
                            Output: n_1.n_nationkey
                            Hash Cond: (n_1.n_regionkey = r_1.r_regionkey)
                            Buffers: shared hit=2
                            -> Seq Scan on public.nation n_1 (cost=0.00..18.10 rows=810 width=8) (actual time=0.010..0.377 rows=50.00 loops=1)
                                Output: n_1.n_nationkey, n_1.n_name, n_1.n_regionkey, n_1.n_comment
                                Buffers: shared hit=1
                            -> Hash (cost=20.62..20.62 rows=4 width=4) (actual time=0.056..0.056 rows=2.00 loops=1)
                                Output: r_1.r_regionkey
                                Buckets: 1024 Batches: 1 Memory Usage: 9kB
                                Buffers: shared hit=1
                                -> Seq Scan on public.region r_1 (cost=0.00..20.62 rows=4 width=4) (actual time=0.021..0.023 rows=2.00 loops=1)
                                    Output: r_1.r_regionkey
                                    Filter: (r_1.r_name = 'EUROPE'::text)
                                    Rows Removed by Filter: 8
                                    Buffers: shared hit=1
                -> Seq Scan on public.nation nation_1 (cost=0.00..18.10 rows=810 width=8) (actual time=0.003..0.005 rows=50.00 loops=6087)
                    Output: nation_1.n_nationkey, nation_1.n_name, nation_1.n_regionkey, nation_1.n_comment
                    Buffers: shared hit=6087
            -> Materialize (cost=0.00..20.64 rows=4 width=4) (actual time=0.000..0.000 rows=2.00 loops=12174)
                Output: region_1.r_regionkey
                Storage: Memory Maximum Storage: 17kB
                Buffers: shared hit=1
            -> Seq Scan on public.region region_1 (cost=0.00..20.62 rows=4 width=4) (actual time=0.013..0.015 rows=2.00 loops=1)
                Output: region_1.r_regionkey
                Filter: (region_1.r_name = 'EUROPE'::text)
                Rows Removed by Filter: 8
                Buffers: shared hit=1

```

```

Planning:
  Buffers: shared hit=240 read=13
  Planning Time: 37.931 ms
  Execution Time: 680918.007 ms
(194 rows)

```

tpch=>